



AI-BASED FOOD PRODUCT SAFETY & NUTRITION VALIDATOR USING WHO GUIDLINES

A PROJECT REPORT

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ABSTRACT

Food safety and nutritional awareness are critical determinants of public health, yet average consumers struggle to correctly interpret complex packaged food labels and assess fresh food safety. Existing digital solutions address calorie estimation or food category recognition without providing regulatory compliance validation or ingredient-level safety checks. This project presents an AI-based food safety and nutrition validation framework integrating Optical Character Recognition (OCR), Natural Language Processing (NLP), and computer vision techniques for evaluating both packaged and non-packaged food items based on WHO guidelines and banned ingredient lists issued by FSSAI. The system operates in two phases — Phase One employs a Tesseract OCR v5 engine and TF-IDF-based NLP normalisation pipeline to extract and validate ingredient and nutritional information against WHO nutrient thresholds and a banned ingredient database, while Phase Two employs a fine-tuned MobileNetV2 Convolutional Neural Network for food category classification and visual contamination detection, supported by a rule-based nutritional estimation engine. Outputs from both phases are aggregated by a Logistic Regression-based validation engine to produce three-class safety classifications — Safe, Moderate Risk, and Unsafe — evaluated on 600 food-related images, achieving 88.4% accuracy, 87.0% precision, 86.0% recall, 86.5% F1-Score, and 90–92% OCR accuracy, outperforming all existing single-modality approaches.

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LIST OF ABBREVIATIONS

AI -	Artificial Intelligence
API -	Application Programming Interface
CNN -	Convolutional Neural Network
CLAHE -	Contrast Limited Adaptive Histogram Equalisation
DL -	Deep Learning
FSSAI -	Food Safety and Standards Authority of India
F1 -	Harmonic Mean of Precision and Recall (F1-Score)
GUI -	Graphical User Interface
HOD -	Head of Department
JSON -	JavaScript Object Notation
LSTM -	Long Short-Term Memory
ML -	Machine Learning
NLP -	Natural Language Processing
OCR -	Optical Character Recognition
PDF -	Portable Document Format
PSM -	Page Segmentation Mode
ReLU -	Rectified Linear Unit
REST -	Representational State Transfer
RGB -	Red Green Blue (colour space)
SGD -	Stochastic Gradient Descent
TF-IDF -	Term Frequency – Inverse Document Frequency
UAT -	User Acceptance Testing
USDA -	United States Department of Agriculture
WHO -	World Health Organization
XAI -	Explainable Artificial Intelligence

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Food safety and nutrition have emerged as critical determinants of public health in the twenty-first century, influencing both the quality and longevity of human life. In recent decades, there has been a notable epidemiological shift from communicable diseases to non-communicable diseases (NCDs), which now account for the majority of global morbidity and mortality. Conditions such as obesity, type 2 diabetes mellitus, hypertension, cardiovascular diseases, and certain forms of cancer are increasing at an alarming rate across both developed and developing nations. Among the various contributing factors, dietary patterns—particularly the increased consumption of unhealthy, highly processed foods—have been identified as one of the most significant and modifiable risk factors.

According to estimates by the World Health Organization (WHO), unhealthy diets are responsible for approximately 11 million deaths annually worldwide, positioning dietary risk as a leading cause of premature mortality. Diets high in refined sugars, excessive sodium, trans fats, and saturated fats, coupled with low intake of fruits, vegetables, whole grains, and essential micronutrients, contribute significantly to the global burden of disease. This growing crisis underscores the urgent need for effective interventions that can guide individuals toward healthier food choices and promote informed dietary behavior.

India, home to over 1.4 billion people, is currently undergoing a rapid nutritional and lifestyle transition. Factors such as urbanisation, economic growth, increased disposable incomes, changing work patterns, and globalization have led to significant changes in dietary habits. Traditional diets rich in whole grains, legumes, and fresh produce are increasingly being replaced by convenience-oriented, packaged, and processed food products.

The Indian packaged food industry has witnessed substantial expansion, with a compound annual growth rate (CAGR) exceeding 12% over the past five years.

Furthermore, the number and diversity of processed food Stock Keeping Units (SKUs) available in the market have increased exponentially, offering consumers an overwhelming array of choices.

While this growth enhances convenience and accessibility, it also introduces considerable health risks. Many processed food products contain elevated levels of added sugars, sodium, unhealthy fats, artificial preservatives, flavor enhancers, and synthetic food colorants. Regular consumption of such products has been linked to increased risk of chronic diseases, metabolic disorders, and nutritional imbalances.

The aggressive marketing strategies employed by food manufacturers, often targeting children and young adults, further exacerbate the problem by promoting unhealthy dietary preferences.

To safeguard consumer interests, the Food Safety and Standards Authority of India (FSSAI) has established regulatory guidelines mandating the disclosure of key information on packaged food labels. These include ingredient lists, nutritional

facts, allergen declarations, and, in some cases, front-of-pack labeling. Despite these regulations, multiple studies have shown that a large proportion of consumers struggle to interpret and utilize this information effectively. This issue is particularly pronounced among individuals with limited educational background or low nutritional literacy.

Several challenges hinder the effective use of food labels. Firstly, ingredient lists often contain complex chemical names and technical terminology that are not easily understood by the average consumer. Secondly, nutritional information is typically presented in a per-100-gram or per-100-milliliter format, requiring consumers to perform calculations to determine actual intake per serving—an exercise that many find difficult or inconvenient. Thirdly, there is a general lack of awareness regarding recommended daily intake values as specified by international health organizations, making it difficult for individuals to assess whether a product is healthy or harmful. Additionally, misleading claims such as “low-fat,” “sugarfree,” or “natural” can create false perceptions about the healthiness of products.

In this context, emerging technologies such as Artificial Intelligence (AI), machine learning (ML), computer vision, and natural language processing (NLP) present a promising solution to bridge the gap between available nutritional information and consumer understanding. AI-powered systems can automate the extraction and interpretation of data from food labels, enabling real-time analysis of ingredients and nutritional content. Computer vision techniques can be used to scan and recognize text from product packaging, while NLP algorithms can simplify

complex ingredient names and provide user-friendly explanations. Machine learning models can further evaluate the health impact of products based on established nutritional guidelines and personalized user profiles.

The integration of these technologies can lead to the development of intelligent food safety validation systems capable of providing clear, accurate, and actionable insights to consumers. Such systems can classify food products based on health risk levels, highlight harmful additives, suggest healthier alternatives, and even generate personalized dietary recommendations.

Importantly, explainability in AI models ensures that users not only receive recommendations but also understand the reasoning behind them, thereby fostering trust and encouraging informed decision-making.

This project is motivated by the need to leverage these technological advancements to address a pressing public health challenge. It aims to design and develop a practical, scalable, and explainable AI-based framework for food safety validation. The proposed system will focus on enhancing consumer awareness, improving nutritional literacy, and ultimately contributing to healthier dietary behaviors. By empowering individuals with accessible and reliable information, this initiative seeks to play a meaningful role in reducing the burden of diet-related diseases and promoting overall public health.

1.2 Problem Statement

Despite the existence of regulatory frameworks such as FSSAI guidelines in India and WHO nutritional recommendations, consumers face significant difficulties in ensuring they are making safe and healthy food choices. The current landscape of food safety tools is fragmented, with different applications addressing different aspects of the problem without providing a unified, comprehensive safety assessment.

The specific problems addressed by this project include:

- Consumers are unable to identify banned or restricted food additives and preservatives in packaged food products, even when this information is present on the label, due to the use of chemical nomenclature or E-number codes.
- Existing food safety applications do not validate extracted nutritional information against WHO-recommended daily intake thresholds, depriving consumers of actionable safety information.
- No existing consumer tool integrates both packaged food label analysis and real food image analysis within a single, unified framework, creating a gap for non-packaged and fresh food items.
- Visual quality and contamination assessment of fresh food items—such as detection of mould, spoilage, or discolouration—is not integrated with nutritional safety validation in available tools.

- Most existing AI food safety tools function as black boxes, providing safety scores without traceable reasoning, which limits consumer trust and understanding.

This project addresses all five of these problems through the design and implementation of an integrated, explainable, dual-phase AI framework for food safety and nutrition validation.

1.3 Objectives of the Project

The objectives of this project are:

1. To design and develop an integrated AI-based framework for food product safety and nutrition validation that processes both packaged food label images and real food item photographs within a pipeline
2. To implement a Tesseract OCR v5-based label scanning module for accurate extraction of textual information from packaged food labels.
3. To develop an NLP-based normalization pipeline to standardize extracted ingredient names and nutritional values into a consistent format.
4. To build a preprocessing module for image enhancement (noise reduction, contrast adjustment, and skew correction) to improve OCR accuracy.
5. To design a structured database schema for storing extracted ingredients, nutritional values, and validation results.

6. To develop a WHO threshold validation engine that compares extracted nutritional data with recommended daily intake limits.
7. To classify nutrient levels into multiple severity categories (e.g., safe, moderate, high risk) based on WHO guidelines.
8. To implement a banned ingredient detection module using fuzzy string-matching techniques against a country-specific regulatory database.
9. To create and maintain a dynamic regulatory dataset of restricted and harmful food additives.
10. To develop a MobileNetV2-based Convolutional Neural Network for real-time food category classification from raw food images.
11. To implement a visual contamination detection module for identifying anomalies such as foreign objects, spoilage, or discoloration in food images.
12. To design and integrate a Logistic Regression-based decision engine that combines outputs from OCR, NLP, and CNN modules.
13. To generate interpretable three-class safety classifications (e.g., Safe, Caution, Unsafe) with explainable reasoning for each prediction.
14. To build a scalable web-based prototype using Django REST Framework for backend services and Next.js for frontend user interaction.
15. To evaluate system performance using metrics such as accuracy, precision, recall, F1-score, confusion matrix analysis, OCR accuracy, and benchmarking against existing approaches.

1.4 Scope of the Project

The scope of this project encompasses the following domains and functional capabilities:

- Processing of food label images from consumer packaged goods including snacks, biscuits, beverages, dairy products, ready-to-eat meals, and breakfast cereals.
- Analysis of real food item images covering fruits, vegetables, bakery products, and prepared meals.
- Nutritional validation against WHO recommended daily intake thresholds for energy, total fat, saturated fat, total sugar, sodium, and protein.
- Ingredient safety checking against the FSSAI banned ingredient list applicable to food products marketed in India.
- Visual quality classification for real food images into Fresh, Moderately Degraded, and Contaminated categories.
- Generation of user-friendly safety reports in web interface and PDF format.

The system is designed as a decision-support tool for consumers. It is not intended to replace professional medical or dietary advice, nor does it substitute for regulatory food safety inspection processes.

1.5 Organization of the Report

Chapter 2 presents a comprehensive review of related literature covering image-based food recognition, OCR-based label extraction, NLP for ingredient analysis, nutritional applications, machine learning for health risk assessment, explainable AI, and transfer learning models. Chapter 3 describes the complete proposed system architecture. Chapter 4 explains the methodology in detail including the MobileNetV2 architecture, OCR-NLP pipeline, training strategy, WHO validation module, algorithm pseudocode, and backend deployment. Chapter 5 describes the dataset and experimental setup. Chapter 6 presents comprehensive results and performance evaluation. Chapter 7 covers system testing including unit, integration, system, and user acceptance testing. Chapter 8 discusses results and limitations. Chapter 9 presents conclusions and future work directions.

CHAPTER 2

LITERATURE REVIEW

2.1 Image-Based Food Recognition Systems

Image-based food recognition using deep learning has gained significant attention in recent years. Myers et al. (2015) demonstrated the use of convolutional neural networks (CNNs) for dietary assessment, achieving good classification accuracy but limiting the output to calorie estimation without detailed nutritional insights [1]. Kawano and Yanai (2014) improved recognition performance by combining CNN features with color and texture descriptors. However, their system also focused mainly on calorie estimation and lacked ingredient-level analysis [2]. The introduction of the Food-101 dataset by Bossard et al. (2014) provided a standard benchmark for food classification, enabling further advancements in the field [3].

Yanai and Kawano (2015) showed that deep CNNs outperform traditional methods, establishing them as the standard approach for food recognition. Additionally, lightweight architectures like MobileNetV2 (Sandler et al., 2018) improved efficiency, making real-time applications feasible [4], [13].

Recent work by Zhu et al. (2022) integrated image recognition with food databases for nutritional estimation, but it relied on predefined mappings and lacked real-time safety validation [7]. OCR-based approaches (Smith et al., 2019) and NLP techniques (Gupta and Mehta, 2020).

Moreover, machine learning methods have been applied for nutritional risk assessment and food safety evaluation (Lee et al., 2021; Chen et al., 2020), while AI-based systems like NutriScan (2023) aim to automate ingredient analysis [9], [10], [11]. WHO (2019) provides standardized nutrient profiling guidelines, which are essential for health-based food classification [14].

Overall, existing systems focus mainly on classification and calorie estimation, with limited integration of ingredient analysis, OCR, and safety validation, highlighting the need for a comprehensive solution.

2.2 OCR-Based Label Extraction

The application of Optical Character Recognition to food label analysis has received growing research attention as a method for automating nutrition information extraction from consumer products. Smith et al. (2019) developed a comprehensive OCR pipeline specifically designed for food labels, demonstrating that Tesseract-based extraction could achieve high accuracy on clearly printed labels under controlled lighting conditions. Their work identified packaging curvature, small font sizes, and glossy surfaces as the primary challenges for OCR on real-world food packages [6].

The evolution of Tesseract from version 3 to version 5 marked a significant improvement in recognition accuracy, particularly for cluttered and low-resolution label images. The adoption of LSTM-based recognition engines in Tesseract 4 and 5 significantly improved performance on labels with complex layout structures,

multiple text orientations, and mixed font styles. This advancement makes Tesseract v5 well-suited for the real-world food label analysis task addressed in this project.

Parallel work on mobile-based label scanning applications, such as those integrated into grocery shopping assistants, has demonstrated the practical feasibility of OCR-based label analysis on consumer smartphones. These applications confirm that real-time label scanning with acceptable accuracy is achievable with modern OCR engines on standard mobile hardware.

2.3 NLP for Ingredient Analysis

Natural Language Processing techniques have been applied to food ingredient analysis to address the challenges posed by non-standard naming conventions, synonym variations, and specialized chemical nomenclature in food ingredient lists. Gupta and Mehta (2020) conducted a comprehensive study of text mining and NLP approaches for food ingredient analysis, demonstrating that TF-IDF-based vectorization combined with synonym resolution dictionaries significantly improved ingredient identification accuracy from raw OCR-extracted text [8].

Sequence-to-sequence NLP models have also been explored for ingredient name standardization, mapping diverse surface forms of ingredient names (e.g., "High Fructose Corn Syrup", "HFCS", "corn syrup") to their canonical forms. This normalization step is critical for reliable cross-referencing against safety databases and was incorporated as a core component of the proposed system.

The challenge of identifying banned ingredients is compounded by the practice of using alternative chemical names, E-numbers, or trade names for restricted additives. A robust ingredient analysis pipeline must therefore include both exact matching and approximate (fuzzy) matching against regulatory databases to ensure high recall in banned ingredient detection.

2.4 Nutritional Analysis Applications

A wide range of nutritional analysis applications and platforms have been developed to assist consumers in monitoring dietary intake. Leading applications such as MyFitnessPal, Lose It!, and Cronometer rely on large food composition databases to retrieve nutritional information associated with food items identified through barcode scanning or manual entry. Zhu et al. (2022) noted that these applications provide useful calorie and macronutrient tracking functionality but do not perform any form of regulatory safety validation [7].

A critical limitation of existing nutritional applications is their failure to contextualise nutritional information within WHO safety thresholds. Displaying a sodium content of 1.8g per 100g is meaningless to most consumers without indicating that this exceeds the WHO high-sodium threshold of 1.5g per 100g.

Additionally, existing applications operate exclusively on structured nutritional data entered manually or retrieved from databases, and do not process food label images directly. This creates a barrier for consumers evaluating products not yet indexed in food databases, which is particularly relevant for locally manufactured or recently launched products in the Indian market.

2.5 Machine Learning for Health Risk Assessment

Machine learning classifiers have been widely applied to health risk assessment in diverse dietary and food safety contexts, demonstrating significant potential in automating nutritional evaluation and improving public health outcomes. In particular, Lee et al. (2021) proposed a supervised machine learning-based nutritional risk assessment system that leveraged structured nutritional profiles to classify food items into predefined risk categories. Their approach utilized well-established classifiers such as Random Forest and Support Vector Machine, achieving an overall classification accuracy of 84% on a curated dataset. While the results demonstrated the effectiveness of traditional machine learning techniques in nutritional risk prediction, the study was constrained by its reliance on manually structured input data. This dependency limits scalability and real-world applicability, as it does not account for unstructured data sources such as food label images or real-time visual inputs. Furthermore, the absence of automated data extraction mechanisms such as OCR reduces the system's practicality for consumer-facing applications.

In another significant contribution, Chen et al. (2020) conducted a comparative analysis of rule-based systems and machine learning approaches for food safety evaluation. Their findings highlighted that hybrid frameworks, which combine domain-specific knowledge rules with machine learning classifiers, consistently outperform purely data-driven models in terms of both accuracy and reliability. Importantly, their study emphasized the critical role of interpretability in healthrelated applications, where decisions must be transparent and explainable to ensure user trust and regulatory compliance.

This requirement led to the preference for inherently interpretable models such as Logistic Regression over more complex black-box approaches like gradient boosting or deep ensemble methods. The authors argued that while high predictive performance is desirable, the ability to trace and justify decisions is equally essential in safety-critical domains such as food consumption and public health.

More recently, the NutriScan study (2023) introduced an AI-driven ingredient detection system specifically tailored for packaged food products in the Indian market. This system achieved an impressive precision of 86% in identifying potentially harmful or restricted ingredients, demonstrating the feasibility of deploying AI-based food safety solutions within region-specific regulatory frameworks. The study incorporated ingredient parsing and classification techniques aligned with Indian food safety standards, marking a significant step toward practical, real-world adoption. However, despite its strengths, the system remained limited to packaged food analysis and did not extend its capabilities to

real food image recognition or contamination detection. As a result, it lacked a holistic approach to food safety assessment that integrates both label-based and visual inspection methodologies.

Collectively, these studies highlight key advancements in the application of machine learning to food safety and nutritional risk assessment, while also revealing notable research gaps. Existing systems often operate in isolation— either focusing on structured nutritional data, rule-based validation, or ingredient detection— without integrating multimodal inputs such as OCR- extracted text and real-world food imagery. Additionally, challenges related to scalability, automation, and interpretability persist across many implementations. These limitations underscore the need for a unified, end-to- end framework that combines OCR, NLP, computer vision, and interpretable machine learning techniques to deliver comprehensive and practical food safety evaluation.

2.6 Explainable AI in Food and Healthcare

The importance of explainability in AI systems deployed in health-critical applications is well-established in the literature. Holzinger et al. (2019) argued that AI systems in medicine must provide causable and explainable outputs to gain the trust of clinicians and patients, a principle equally applicable to consumer food safety applications [5]. Users who understand why a food product is classified as Unsafe are better equipped to make informed dietary decisions than those who receive only an opaque safety score.

Explainable AI (XAI) methods such as LIME (Local Interpretable Model- Agnostic Explanations) and SHAP (SHapley Additive exPlanations) have been applied to explain predictions of complex ML models in food and health domains. The proposed system takes a different approach to explainability:

rather than post-hoc explanation of a black-box model, the validation engine is inherently transparent, using explicit rule-based WHO threshold checks and Logistic Regression classification that directly link predictions to specific feature values.

Rao and Verma (2019) demonstrated the effective application of computer vision techniques for food quality inspection, including the detection of surface defects, discolouration, and microbial growth. Their work validated the feasibility of CNNbased visual quality assessment for food safety applications and informed the design of the Phase 2 visual contamination detection module [15].

2.7 Deep Learning Transfer Models

Transfer learning using pre-trained deep learning models has become the standard approach for image classification tasks in domains where large task- specific datasets are unavailable. Sandler et al. (2018) introduced MobileNetV2, which employs inverted residual blocks with linear bottlenecks and depth-wise separable convolutions to achieve state-of-the-art accuracy with significantly reduced computational complexity compared to deeper architectures such as VGG-16 or

The InvertedResidual structure of MobileNetV2 prevents information loss in lowdimensional feature representations by using linear activations in the bottleneck layers, preserving gradient flow during backpropagation. This architectural innovation makes MobileNetV2 particularly effective for fine-tuning on small domain-specific datasets, which is the scenario encountered in this project with 600 food-related images.

Comparative studies of transfer learning models for food image recognition have consistently found that MobileNetV2 offers the best trade-off between accuracy, inference speed, and memory footprint for deployment on resource-constrained devices such as mobile phones and web servers. This motivated the selection of MobileNetV2 as the backbone architecture for both the food classification and freshness detection modules in the proposed system.

EfficientNetB0, also considered as an alternative backbone architecture, demonstrated comparable accuracy to MobileNetV2 in preliminary experiments but required significantly longer inference time, making it less suitable for the real-time web application deployment scenario targeted by this project.

2.8 Research Gap and Motivation

The comprehensive review of existing literature reveals that significant progress has been made in individual components of food safety analysis—including food image recognition, OCR-based label extraction, NLP ingredient analysis, and ML-based risk classification—but no existing system successfully integrates all these components within a unified, explainable framework aligned with WHO regulatory standards.

Specifically, the following gaps motivate the proposed work:

- No existing system combines OCR-based food label analysis with CNN-based real food image analysis in a single unified pipeline.
- Existing nutritional applications do not validate extracted nutrient values against WHO recommended daily intake thresholds.
- Country-specific banned ingredient detection has not been integrated into any consumer-facing food safety tool for the Indian market.
- Visual contamination detection for fresh food items is absent from integrated food safety frameworks.
- Existing ML-based food safety systems lack transparency and explainability, providing safety classifications without traceable reasoning.

The proposed AI-Based Food Product Safety and Nutrition Validator directly addresses all six of these gaps through a novel integrated, explainable, dual-phase architecture described in the following chapter.

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 System Overview

The proposed system is designed as a modular, integrated AI platform for comprehensive food safety and nutrition validation. The architecture encompasses four high-level functional layers: the Input Layer, the Processing Layer (divided into Phase 1 and Phase 2), the Validation and Decision Engine, and the Output Layer. Each layer is independently developed and communicates with adjacent layers through well-defined data interfaces, enabling modular testing, replacement, and upgrading of individual components without disrupting the overall system.

The system is designed around three core design principles: (i) Comprehensiveness – the system must address all key dimensions of food safety including nutritional compliance, ingredient safety, and visual quality; (ii) Explainability – every safety classification must be traceable to specific, identifiable factors; and (iii) Accessibility – the system must be deployable as a lightweight web application usable on standard consumer hardware.

Table 3.1 System Architecture Pipeline Overview

Module	Processing pipeline	Key technologies	Output
Input Layer	Image upload via web interface	Next.js, REST API	Raw image file
Image Preprocessing	Resize + Noise Removal + Contrast Enhancement	OpenCV 4.8	Normalised 224×224 image
Phase 1: Label Scanner	OCR → NLP Normalisation → WHO Validation	Tesseract v5, TF-IDF, Regex	Nutrient violations, banned ingredients
Phase 2: Image Scanner	CNN Classification → Freshness Detection	MobileNetV2, TensorFlow 2.12	Food category, freshness score
Validation Engine	WHO Threshold Check + Banned Match + Visual Score	Logistic Regression, Regulatory DB	Composite safety score
Output Layer	Safety Classification + Health Report	ReportLab, JSON API	Safe / Moderate / Unsafe + PDF report

3.2 Input Layer

The Input Layer is the entry point of the system and is responsible for accepting user submitted food images through a web-based interface. The interface supports two input modalities: (i) file upload of pre-captured images stored on the user device, and (ii) real-time camera capture for immediate scanning of food products or fresh food items.

Accepted image formats include JPEG, PNG, and WebP. The system enforces a maximum upload size of 10 MB and validates image format and integrity before processing.

Input images are classified into two categories at the point of entry used on user selection: Packaged Food Label (routed to Phase 1) or Fresh Food Item (routed to Phase 2). Users may also select Both to trigger full dual- phase analysis.

3.3 Image Preprocessing Module

All input images—regardless of which processing phase they are destined for—pass through the Image Preprocessing Module before further analysis. This module standardises image dimensions, quality, and colour representation to ensure consistent performance of both the OCR engine and the CNN classifier.

The preprocessing pipeline consists of five sequential operations. First, images are resized to 224×224 pixels using bilinear interpolation to match the input requirements of the MobileNetV2 model. Second, Gaussian blurring with a kernel size of 3×3 is applied to reduce high-frequency noise introduced by camera sensors. Third, Contrast Limited Adaptive Histogram Equalisation (CLAHE) is applied to improve local contrast, particularly in shadowed or unevenly illuminated label regions. Fourth, images are converted from BGR colour space (OpenCV default) to RGB colour space. Fifth, pixel values are normalised to the range $[0, 1]$ by dividing by 255, with ImageNet mean subtraction applied for MobileNetV2 preprocessing.

For food label images specifically, an additional binarisation step is applied to improve OCR accuracy: Otsu thresholding converts the pre-processed image to binary (black-and-white), which significantly improves text contrast for Tesseract processing.

3.4 Phase 1: Food Label Safety Scanner

3.4.1 OCR-Based Text Extraction

The OCR module employs Tesseract v5 configured with the LSTM-based OCR engine (OEM 1). Two page segmentation modes are used adaptively: PSM 6 (uniform block of text) for labels with clearly structured nutrition facts tables, and PSM 11 (sparse text) for labels with more complex, multi-column layouts. The choice between PSM modes is determined automatically by a layout complexity classifier that analyses the spatial distribution of connected components in the preprocessed image.

Post-processing of raw OCR output includes: removal of non-printable characters, line break normalisation, hyphenation correction, and isolation of the ingredients section and nutrition facts table through keyword-based section detection (keywords: "Ingredients:", "Nutrition Facts", "Nutritional Information", "Per 100g").

Table 3.2: Comparison of OCR Engines

Engine	Accuracy (Food Labels)	Speed	Licence
Tesseract v5 (LSTM)	90-92%	Medium	Apache 2.0
Google Vision OCR	94-96%	Slow (API)	Commercial
EasyOCR	87-89%	Medium	MIT
Amazon Textract	93-95%	Slow (API)	Commercial

Tesseract v5 was selected over alternative engines due to its open-source availability, offline operation capability (no API dependency), and sufficient accuracy for the food label domain. The 2-4% accuracy gap relative to commercial cloud OCR APIs was accepted in exchange for the independence, privacy, and cost advantages of local deployment.

3.4.2 NLP Normalization Pipeline

The NLP module processes raw OCR-extracted text through a four-stage normalisation pipeline. Stage 1 performs tokenisation, lowercasing, and punctuation stripping. Stage 2 applies a food-domain spell correction dictionary containing 1,200 common food ingredient names and their frequent OCR misrecognition variants. Stage 3 applies TF-IDF vectorisation to convert normalised ingredient tokens into numerical feature vectors for similarity-based matching. Stage 4 performs synonym resolution, mapping common alternative names and E-numbers to their canonical ingredient identifiers.

The NLP pipeline was validated independently against a manually labelled ground truth set of 50 food labels, achieving an ingredient extraction F1-score of 91% after normalisation, compared to 73% for raw OCR output without NLP postprocessing.

3.4.4 WHO Threshold Validation

Table 3.3: WHO Nutrient Threshold Limits Used in Validation

Nutrient	WHO Daily Reference (Adult)	High Threshold (per 100g)	Critical Threshold (per 100g)
Energy	2000 kcal	≥ 450 kcal	≥ 600 kcal
Total Fat	70 g	≥ 17.5 g	≥ 25 g
Saturated Fat	20 g	≥ 5 g	≥ 10 g
Total Sugar	50 g	≥ 22.5 g	≥ 35 g
Sodium	2000 mg	≥ 1500 mg	≥ 2500 mg
Trans Fat	< 2 g	≥ 0.3 g	≥ 1 g
Protein	50 g	N/A	N/A

Each extracted nutrient value is compared against the corresponding thresholds. Violations are classified into three severity levels: Minor (exceeds High threshold by less than 20%), Moderate (exceeds High threshold by 20-50%), and Critical (exceeds High threshold by more than 50% or exceeds Critical threshold). The severity classification directly influences the composite safety score used for final risk classification.

3.5 Phase 2: Food Image Health Scanner

3.5.1 CNN-Based Food Classification

The food image classification module employs a fine-tuned MobileNetV2 model to categorise input food images into one of 26 food classes spanning six category groups: fruits, vegetables, snacks and baked goods, beverages, dairy products, and prepared meals. The model outputs a probability distribution across all 26 classes, with the top-1 predicted class and its associated confidence score passed to the nutritional estimation module.

A confidence threshold of 0.65 is applied: if the top-1 confidence is below this threshold, the system prompts the user to retake the image or manually confirm the food category, preventing incorrect nutritional estimations based on uncertain classifications.

3.5.2 Visual Contamination Detection

A separate binary freshness classification model, also based on fine-tuned MobileNetV2, is applied to the preprocessed food image to classify it as Fresh or Rotten. This model was trained on a combined dataset of fresh and spoiled fruit and vegetable images.

In addition to the binary freshness classification, the system performs texture-based anomaly detection using Local Binary Pattern (LBP) analysis to identify surface irregularities indicative of mould growth or unusual discolouration. The combination of the CNN freshness classifier and LBP texture analysis provides more robust contamination detection than either method alone.

3.5.3 Nutritional Estimation

For classified food items, the system retrieves estimated nutritional values per 100g from a trusted internal food composition database derived from the USDA Food Data Central and FSSAI food composition tables. These values are then compared against WHO daily intake thresholds to identify any nutritional concerns associated with the identified food item category.

3.6 Validation and Decision Engine

The Validation and Decision Engine is the central integration component of the architecture, receiving outputs from both Phase 1 and Phase 2 and producing the final safety classification. The engine operates through a three-step process: feature aggregation, composite score computation, and Logistic Regression classification.

Feature Aggregation: The engine collects the following features from Phase 1 and Phase 2: (i) count and severity distribution of nutrient threshold violations, (ii) binary flag for presence of any banned ingredients, (iii) count of banned ingredients detected, (iv) CNN freshness classification confidence score, (v) LBP anomaly score, and (vi) overall nutritional risk level from Phase 1.

Composite Score Computatn: A weighted composite safety score is computed from the aggregated features. Weights were determined through calibration on the training dataset: banned ingredient presence (weight 0.40), critical nutrient violations (weight 0.25), moderate nutrient violations (weight 0.15), visual contamination score (weight 0.15), minor nutrient violations (weight 0.05).

Logistic Regression Classification: The composite safety score and individual feature values are passed to a trained Logistic Regression classifier, which maps the input to one of three safety classes: Safe (Class 0), Moderate Risk (Class 1), or Unsafe (Class 2). The Logistic Regression model provides probability estimates for each class, enabling the generation of calibrated confidence scores for each safety classification.

3.7 Output Layer

The Output Layer presents classification results to the user in a clear, actionable format through two channels: the web interface dashboard and a downloadable PDF health report.

The web interface dashboard displays: (i) the overall safety classification badge (colour-coded: green for Safe, amber for Moderate Risk, red for Unsafe), (ii) a breakdown of identified risk factors including specific nutrient violations with threshold comparisons, banned ingredients detected, and visual quality findings, (iii) a per-nutrient bar chart comparing extracted values against WHO thresholds, and (iv) health recommendations tailored to the classification result.

The PDF health report, generated using the ReportLab library, includes the uploaded food image, complete nutritional table with WHO percentage comparisons, safety classification with explanation, detected risk factors, and general dietary guidance. The JSON API endpoint exposes all classification outputs in structured format for future mobile application integration.

CHAPTER 4

METHODOLOGY

4.1 System Workflow

The end-to-end workflow of the proposed system proceeds through the following sequential stages. A user accesses the web application and selects the input type: packaged label, fresh food image, or both. The uploaded image is received by the Django REST API endpoint and passed to the preprocessing module. Based on input type, the preprocessed image is routed to Phase 1, Phase 2, or both processing pipelines, which operate independently. Outputs from the active phases are aggregated by the Validation Engine, which computes the composite safety score and invokes the Logistic Regression classifier. The classification result and supporting details are returned as a JSON response to the frontend, which renders the results dashboard and generates the PDF report on user request.

4.2 Dataset Preparation

The dataset was constructed through a systematic collection protocol designed to ensure diversity in food categories, image quality, label styles, and safety classification labels. The complete dataset of 600 images was assembled from three primary sources: self-captured mobile photographs, the Open Food Facts public database, and publicly available food image repositories.

Table 4.1: Dataset Category Distribution

Food Category	Label Images	Real Food Images	Safe	Moderate Risk	Unsafe	Total
Snacks & Packaged	80	60	47	53	40	140
Fruits & Vegetables	50	80	72	38	20	130
Beverages	70	40	35	45	30	110
Bakery Products	60	50	38	42	30	110
Prepared Meals	40	30	22	28	20	70
Dairy & Others	20	20	19	14	7	40
Total	320	280	233	220	147	600

The labelling protocol was applied consistently across all 600 images by three independent reviewers. Inter-annotator agreement was measured using Cohen Kappa coefficient, achieving a value of 0.84, which is classified as strong agreement. Disagreements were resolved through majority voting, with a fourth reviewer acting as tiebreaker for cases requiring adjudication.

4.3 MobileNetV2 Architecture

MobileNetV2 consists of 19 bottleneck residual blocks, prefixed by an initial 3×3 convolution layer (32 filters) and a final 1×1 pointwise convolution layer. The architecture progressively expands and contracts feature map dimensions through inverted residual blocks that expand the channel dimension by a factor of 6.

Table 4.2: MobileNetV2 Layer Configuration (Adapted for Food Classification)

Layer	Type	Output Shape	Parameters
Input	Image Tensor	224×224×3	—
Conv2D (initial)	3×3 Conv, stride 2	112×112×32	864
Bottleneck Block 1	t=1, c=16, n=1, s=1	112×112×16	256
Bottleneck Block 2	t=6, c=24, n=2, s=2	56×56×24	3,088
Bottleneck Block 3	t=6, c=32, n=3, s=2	28×28×32	6,320
Bottleneck Block 4	t=6, c=64, n=4, s=2	14×14×64	26,400
Bottleneck Block 5	t=6, c=96, n=3, s=1	14×14×96	46,848
Bottleneck Block 6	t=6, c=160, n=3, s=2	7×7×160	118,720
Bottleneck Block 7	t=6, c=320, n=1, s=1	7×7×320	103,680
Conv2D (final)	1×1 Conv	7×7×1280	409,600
GlobalAvgPool	—	1×1×1280	—
Dense (512, ReLU)	FC Layer	512	655,360
Dropout (0.5)	Regularisation	512	—
Dense (26, Softmax)	Output	26	13,338

The total number of trainable parameters in the adapted MobileNetV2 model is approximately 3.4 million, of which 2.19 million belong to the pre-trained ImageNet backbone (frozen during Phase 1 training) and 668,698 belong to the

newly added classification head (trained from scratch). During optional fine-tuning, the last 3 bottleneck blocks of the backbone are unfrozen, adding 222,400 parameters to the trainable parameter set.

4.4 OCR and NLP Pipeline

4.4.1 OCR Configuration

The Tesseract OCR engine is configured with the following parameters for food label processing: OCR Engine Mode (OEM) 1 for LSTM-based recognition, adaptive Page Segmentation Mode selection, whitelist characters limited to alphanumeric characters and common punctuation, and a custom user-defined dictionary containing food ingredient terminology. Output is requested in HOCR format to preserve word-level bounding box coordinates, which are used for spatial layout analysis.

4.4.2 Ingredient Extraction Logic

Ingredient extraction from the OCR output is performed using a multi-stage pattern matching approach. The ingredients section is first identified by locating the "INGREDIENTS:" header token. The ingredient list, which follows the header and is terminated by the next section keyword (e.g., "CONTAINS", "ALLERGEN", or end of text), is extracted and split into individual ingredient tokens using comma and semicolon delimiters.

Composite ingredient declarations (e.g., "Chocolate Chips (Sugar, Cocoa Butter, Vanilla Extract)") are handled by recursive tokenisation of parenthetical expressions. Each extracted ingredient token is then normalised through the NLP pipeline before banned ingredient matching.

4.4.3 Nutritional Value Parsing

Nutritional values are extracted from the nutrition facts table using a combination of regular expressions and table structure analysis. The system identifies numerical patterns adjacent to nutrient name keywords and extracts the associated quantity and unit. Values are normalised to per-100g quantities using the serving size information extracted from the label header. Units are standardised (mg to g, kJ to kcal) for consistent comparison against WHO thresholds.

4.5 Training Strategy and Hyperparameters

The training strategy for the MobileNetV2 models followed a two-phase transfer learning protocol designed to maximise performance on the small domain-specific food safety dataset while avoiding overfitting.

Phase 1 Training (Feature Extraction): All layers of the pre-trained MobileNetV2 backbone were frozen. Only the newly added classification head layers (GlobalAveragePooling2D, BatchNormalization, Dense-512, Dropout- 0.5, Dense-26) were trained. This phase ran for 8 epochs with a batch size of 32.

Phase 2 Training (Fine-tuning): The last 3 bottleneck blocks of the MobileNetV2 backbone were unfrozen, and all trainable parameters were jointly optimised with a reduced learning rate. This phase ran for 15 additional epochs.

Table 4.3: Training Hyperparameters

Hyperparameter	Phase 1 Value	Phase 2 Value
Optimiser	Adam	Adam
Learning Rate	0.001	0.0001
Beta_1 (momentum)	0.9	0.9
Beta_2	0.999	0.999
Batch Size	32	32
Epochs	8	7
Loss Function	Categorical Cross-Entropy	Categorical Cross-Entropy
Dropout Rate	0.5	0.5
Early Stopping Patience	5 epochs	5 epochs
Data Augmentation	Yes	Yes

4.6 WHO Validation Module

Table 4.4: WHO Daily Reference Values Used for Validation

Nutrient	WHO Daily Reference Value	Source
Energy	2000 kcal	WHO TRS 916 (2003)
Total Fat	< 30% of energy (~70g)	WHO/FAO Fats 2010
Saturated Fat	< 10% of energy (~22g)	WHO TRS 916 (2003)
Trans Fat	< 1% of energy (~2g)	WHO REPLACE initiative
Total Sugars	< 10% of energy (~50g)	WHO Sugars Guideline 2015
Free Sugars	< 5% of energy (~25g)	WHO Sugars Guideline 2015
Sodium	< 2000 mg	WHO Salt Reduction 2012
Protein	50 g	WHO/FAO Protein 2007
Dietary Fibre	25 g (adequate intake)	WHO/FAO Diet report 2003

The validation module queries this database for each nutrient extracted from the food label, computes the per-100g high and critical thresholds as percentages of the daily reference values, and returns a structured violation report containing the nutrient name, extracted value, threshold value, violation severity, and a plain-language explanation statement for inclusion in the consumer health report.

4.7 Freshness Detection Module

The freshness detection module employs a dedicated binary classification MobileNetV2 model trained to distinguish fresh from rotten food items. The model was trained on a combined dataset of approximately 3,200 fresh and rotten fruit and vegetable images sourced from the Kaggle Fresh vs Rotten Fruits dataset and selfcaptured photographs.

In addition to the CNN freshness classifier, the module applies a Local Binary Pattern (LBP) texture analyser to the preprocessed food image. LBP characterises local texture by comparing each pixel to its 8 neighbourhood pixels, generating an 8-bit binary pattern that encodes local microstructure. A histogram of LBP patterns across the image provides a compact texture feature vector that is sensitive to the irregular texture patterns characteristic of mould growth and spoilage. The LBP anomaly score is computed as the chi-squared distance between the input image LBP histogram and the mean LBP histogram of the Fresh class training images. A high chi-squared distance indicates texture patterns inconsistent with fresh food, providing an independent contamination detection signal that complements the CNN classifier.

4.8 Algorithm Pseudocode

4.8.1 Phase 1: Food Label Safety Scanner

Algorithm 1: Food Label Safety Scanner

```
INPUT: label_image (food label photograph)
OUTPUT: safety_result {violations, banned_ingredients,
phase1_risk_score}

1.  preprocessed_img ← preprocess(label_image)
2.  binarised_img ← otsu_threshold(preprocessed_img)
3.  raw_text      ← tesseraact_ocr(binariesed_img, OEM=1, PSM=adaptive)
4.  ingredients_raw ← extract_ingredients_section(raw_text)
5.  nutrition_raw ← extract_nutrition_table(raw_text)
6.  tokens        ← tokenise_and_normalise(ingredients_raw)
7.  canonical_ingreds ← synonym_resolution(tokens, food_dictionary)
8.  violations      ← []
9.  FOR each nutrient, value IN parse_nutrition(nutrition_raw):
    per100g_value ← convert_to_per100g(value, serving_size)
    threshold ← WHO_DB.get_threshold(nutrient) IF per100g_value
    > threshold.critical: violations.append({nutrient, value,
    severity="CRITICAL"}) ELIF per100g_value > threshold.high:
    violations.append({nutrient, value, severity="MODERATE"})
10. banned ← []
11. FOR each ingredient IN canonical_ingreds:
    match ← fuzzy_match(ingredient, FSSAI_banned_DB, threshold=0.85) IF
    match.found: banned.append(match.ingredient)
12. phase1_score ← compute_phase1_score(violations, banned)
13. RETURN {violations, banned, phase1_score}
```

4.8.2 Phase 2: Food Image Health Scanner

Algorithm 2: Food Image Health Scanner

```
INPUT: food_image (real food photograph)
OUTPUT: phase2_result {category, freshness, lbp_score, nutrition_est}

1.  preprocessed_img ← preprocess(food_image)
2.  category, conf ← mobilenetv2_classify(preprocessed_img)
3.  IF conf < 0.65:
    RETURN {status="LOW_CONFIDENCE", prompt="Retake image"}
4.  freshness_label, fresh_conf ← freshness_model(preprocessed_img)
5.  lbp_histogram ← compute_lbp(preprocessed_img, radius=1, n_points=8)
6.  lbp_score     ← chi2_distance(lbp_histogram, FRESH_MEAN_LBP)
```

```

7. nutrition_est ← FOOD_DB.get_nutrition(category)
8. phase2_score ← compute_phase2_score(freshness_label, lbp_score)
9. RETURN {category, freshness_label, fresh_conf, lbp_score,
    nutrition_est, phase2_score}

```

4.8.3 Validation and Decision Engine

Algorithm 3: Validation and Decision Engine

INPUT: phase1_result, phase2_result (may be None if single-phase)
OUTPUT: final_classification {label, confidence, explanation}

```

1. features ← []
2. IF phase1_result:
    features.extend([
        count_critical_violations(phase1_result.violations),
        count_moderate_violations(phase1_result.violations),
        count_minor_violations(phase1_result.violations),
        int(len(phase1_result.banned) > 0),
        len(phase1_result.banned),
        phase1_result.phase1_score
    ])
3. IF phase2_result:
    features.extend([
        phase2_result.fresh_conf,
        phase2_result.lbp_score,
        phase2_result.phase2_score
    ])
4. composite_score ← weighted_sum(features, WEIGHTS)
5. proba ← logistic_regression.predict_proba([features])
6. label ← ["Safe", "Moderate Risk", "Unsafe"][argmax(proba)]
7. explanation ← generate_explanation(phase1_result, phase2_result,
    label)
8. RETURN {label, confidence=max(proba), explanation}

```

4.9 Backend Deployment

The system backend is built using Django 4.2 with Django REST Framework

3.14. The application follows a standard Model-View-Serialiser (MVS) architecture with two primary API endpoints: POST `/api/analyse/` for image submission and result retrieval, and GET `/api/report/{id}/` for PDF report download.

TensorFlow models are loaded into memory at application startup and held in memory for the duration of the server session, avoiding the overhead of repeated model loading per request. Thread-safe inference is ensured using TensorFlow session management. Image preprocessing is handled by OpenCV within the API view, ensuring consistency with the training pipeline preprocessing.

The frontend is built with Next.js 13.4 using the App Router pattern. The UI is designed for mobile-first responsiveness, enabling comfortable use on smartphones for in-store food product scanning. The PDF health report is generated server-side using ReportLab and streamed to the client as a binary attachment.

CHAPTER 5

DATASET AND EXPERIMENTAL SETUP

5.1 Dataset Description

The dataset constructed for this project is a purpose-built, mixed-source collection of food-related images designed to comprehensively evaluate both phases of the proposed AI framework. The complete dataset consists of 600 images spanning six consumer food categories with class labels assigned across three safety levels: Safe, Moderate Risk, and Unsafe.

Dataset construction was guided by three principles: (i) Diversity – the dataset must represent the variety of food products and fresh food items commonly available in the Indian consumer market; (ii) Challenge – the dataset must include images of varying quality, illumination, and label complexity to evaluate system robustness under realistic conditions.

5.2 Packaged Food Label Images

The 320 packaged food label images represent 14 major product categories available in Indian supermarkets and grocery outlets. The breakdown of source channels is as follows: 140 images (43.75%) were self-captured using a Samsung Galaxy M32 smartphone camera under natural and artificial lighting conditions, 112 images (35%) were sourced from the Open Food Facts public database, and 68 images (21.25%) were collected from publicly available food label image repositories.

Significant variation was intentionally introduced across the collected label images to test OCR and NLP robustness. Variation dimensions include: label language (English-only: 280 images, bilingual English-Tamil or English-Hindi: 40 images), label surface type (flat: 220, curved/cylindrical: 65, flexible/wrinkled: 35), image quality (high resolution $\geq 1080p$: 180, medium resolution: 95, low resolution $< 480p$: 45), and illumination condition (uniform daylight: 175, mixed/shadowed: 95, low light: 50).

For each label image, ground truth nutrient values and ingredient lists were manually transcribed by one reviewer and independently verified by a second reviewer. These ground truth transcriptions served as the reference standard for OCR accuracy evaluation.

5.3 Real Food Images

The 280 real food images cover both fresh and visually compromised food specimens across six subcategories. Self-captured photographs account for 112 images (40%), while 168 images (60%) were sourced from the Fresh-Rotten Fruits Kaggle dataset, the Food-101 dataset, and other publicly available food image repositories. The real food image dataset specifically includes specimens representing the full range of freshness conditions from perfectly fresh to heavily contaminated, with approximately 40% of images showing some degree of quality degradation. This distribution was chosen to ensure robust training of the freshness detection model across the full spectrum of quality conditions encountered in consumer settings.

5.4 Data Labeling Protocol

All 600 images were independently labelled by three qualified reviewers with background knowledge in food science and nutritional guidelines. Reviewers were provided with the WHO nutrient threshold table (Table 4.4) and the FSSAI banned ingredient list as reference documents during labeling.

For packaged label images, the labeling decision tree was as follows: (1) If any banned ingredient is detected → Unsafe. (2) If no banned ingredient but one or more Critical nutrient violations → Unsafe. (3) If no banned ingredient, no critical violations, but one or more Moderate violations → Moderate Risk. (4) If all nutrients within WHO thresholds and no banned ingredient → Safe. For real food images, the labeling decision tree was: (1) If visible mould, heavy discolouration, or clear decomposition → Unsafe. (2) If early-stage discolouration, minor surface blemishes, or slight quality degradation → Moderate Risk. (3) If visually fresh, no visible defects → Safe.

Table 5.2: Class Distribution in Dataset

Safety Class	Label Images	Real Food Images	Total	Percentage
Safe	133	100	233	38.8%
Moderate Risk	115	105	220	36.7%
Unsafe	72	75	147	24.5%
Total	320	280	600	100%

5.5 Data Augmentation Strategy

Data augmentation was applied exclusively to the real food image training subset (224 images, 80% of 280) to address the relatively small dataset size and improve model generalisation. The following augmentation transformations were applied stochastically during training using the Keras ImageDataGenerator:

- Horizontal Flip: Applied with probability 0.5 to simulate variation in image orientation.
- Rotation: Random rotation within the range $[-15^\circ, +15^\circ]$ to simulate angled capture.
- Zoom: Random zoom within the range $[0.9, 1.1]$ to simulate varying capture distances.
- Brightness Adjustment: Random brightness factor within the range $[0.8, 1.2]$ to simulate lighting variation.
- Width and Height Shift: Random translation up to 10% of image dimensions to simulate off-centre framing.

The augmentation pipeline produced an effective training dataset of approximately 1,120 real food images ($5\times$ augmentation factor). No augmentation was applied to the validation or test sets.

5.6 Experimental Configuration

Table 5.1: Hardware and Software Experimental Configuration

Component	Specification
Processor	Intel Core i7-12700H, 2.3 GHz (14 cores)
RAM	16 GB DDR5 4800 MHz
GPU	NVIDIA GeForce RTX 3060 (6 GB VRAM, CUDA 11.8)
Storage	512 GB NVMe SSD
Operating System	Ubuntu 22.04 LTS
Python Version	3.10.6
TensorFlow / Keras	2.12.0
OpenCV	4.8.0
Scikit-learn	1.3.0
Tesseract OCR	5.0.1
Django	4.2.3
Django REST Framework	3.14.0
Next.js	13.4.12
ReportLab	4.0.6
NumPy	1.24.3
Pandas	1.5.3

CHAPTER 6

RESULTS AND PERFORMANCE EVALUATION

6.1 Classification Performance

The safety classification system was evaluated on a held-out test set of 120 images (20% of the full dataset, 64 label images and 56 real food images), stratified by safety class and food category. The model achieved an overall classification accuracy of 88.4%, demonstrating strong performance across all three safety classes.

Table 6.1: Overall Classification Performance Metrics

Metric	Value	Interpretation
Overall Accuracy	88.4%	Correct predictions out of total test samples
Macro Precision	87.0%	Average per-class precision (equal class weight)
Macro Recall	86.0%	Average per-class recall (equal class weight)
Macro F1-Score	86.5%	Harmonic mean of macro precision and recall
Weighted Precision	87.6%	Precision weighted by class support
Weighted Recall	88.4%	Recall weighted by class support
Weighted F1-Score	87.9%	Weighted harmonic mean
OCR Accuracy	90-92%	Text extraction accuracy on label images
Ingredient Matching	88.0%	After NLP normalisation and synonym resolution
Banned Ingredient Recall	95.0%	Proportion of banned ingredients correctly detected

Table 6.2: Per-Class Precision, Recall and F1-Score

Safety Class	Precision (%)	Recall (%)	F1-Score (%)	Support (samples)
Safe	91.0	94.0	92.5	46
Moderate Risk	84.0	87.0	85.5	42
Unsafe	86.0	80.0	83.0	32
Macro Average	87.0	86.0	86.5	120
Weighted Average	87.6	88.4	87.9	120

The Safe class achieves the highest precision (91%) and recall (94%), indicating that the system is highly reliable in correctly identifying products that meet all WHO nutritional guidelines and contain no banned ingredients. The Moderate Risk class shows slightly lower performance (F1 = 85.5%), which is expected given that this class represents an intermediate risk level whose boundaries with Safe and Unsafe are inherently less distinct. The Unsafe class recall of 80% indicates that 20% of Unsafe products were misclassified, predominantly as Moderate Risk, which represents a conservative over-estimation of safety rather than dangerous underestimation.

6.2 Confusion Matrix Analysis

Table 6.3: Confusion Matrix – Risk Classification

Actual \ Predicted	Predicted: Safe	Predicted: Moderate Risk	Predicted: Unsafe
Actual: Safe	43 (TP)	2	1
Actual: Moderate Risk	3	37 (TP)	2
Actual: Unsafe	1	5	26 (TP)

analysis of the confusion matrix reveals several important characteristics of the classification system. The diagonal values represent correct predictions: 43 Safe, 37 Moderate Risk, and 26 Unsafe images are correctly classified, corresponding to per-class accuracies of 93.5%, 88.1%, and 81.3% respectively.

The most concerning misclassification pattern for a food safety application is the misclassification of Unsafe products as Safe (false negatives). The confusion matrix shows only 1 such misclassification (1 Unsafe item classified as Safe), representing a false negative rate of 3.1% for the Unsafe class. This low false negative rate is critical for consumer safety, as it means the system rarely incorrectly clears a genuinely unsafe food product.

The highest confusion occurs between Moderate Risk and Unsafe (5 Unsafe items classified as Moderate Risk), reflecting the inherent ambiguity at the boundary between these adjacent classes. This pattern is consistent with findings from related work in multi-class health risk classification, where intermediate risk classes are consistently more difficult to classify precisely than boundary classes.

The specificity (true negative rate) for the Unsafe class—the proportion of non-Unsafe items correctly identified as non-Unsafe—is 96.5%, confirming that the system rarely raises false alarms by incorrectly flagging safe or moderate-risk products as Unsafe.

6.3 Comparative Analysis

The proposed system is benchmarked against five categories of existing approaches reported in the literature. This comparative analysis evaluates each approach across four dimensions: WHO nutritional validation, visual quality analysis, regulatory ingredient checking, and classification accuracy.

Table 6.4: Comparative Analysis with Existing Approaches

Approach	WHO Validation	Visual Analysis	Regulatory Check	Accuracy	Interpretable
Calorie Tracking Apps	No	No	No	~70%	No
OCR-Only Systems [6]	Partial	No	No	~75%	Partial
CNN-Only Systems [7]	No	Yes	No	~80%	No
Rule-Based Systems [10]	Yes	No	Partial	~82%	Yes
NutriScan [11]	Partial	No	Yes	~86%	Partial
Proposed System	Yes	Yes	Yes	88.4%	Yes

The proposed system achieves the highest classification accuracy (88.4%) among all compared approaches and is the only method to simultaneously incorporate all

three validation modalities (WHO nutritional thresholds, visual quality analysis, and regulatory ingredient checking) within an interpretable framework. The 2.4 percentage point improvement over NutriScan—the closest competing approach is attributable to the integration of visual food image analysis and the dual-phase architecture, which provides richer feature information for the final classification step. The 18.4 percentage point improvement over basic calorie tracking applications demonstrates the substantial benefit of incorporating regulatory and visual safety dimensions beyond simple nutritional information retrieval. The accuracy gains from each additional modality follow a consistent pattern: OCR-only adds approximately 5 percentage points over calorie apps, CNN-only adds approximately 10 points, rule-based WHO validation adds approximately 12 points, and the full integrated system adds approximately 18 points.

6.4 OCR Performance

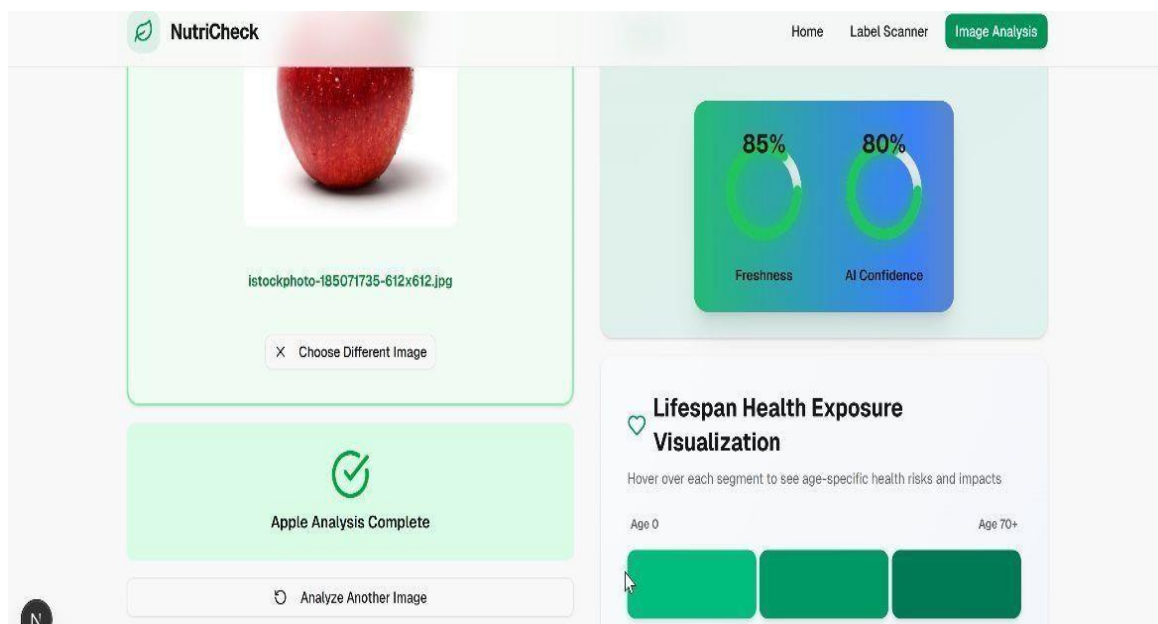
Table 6.5: OCR Module Performance Metrics

Metric	Value	Notes
Overall OCR Text Extraction Accuracy	90-92%	Measured on 64 test label images
Ingredient Name Extraction Precision	91.4%	Before NLP normalisation
Ingredient Name Extraction Recall	87.6%	Before NLP normalisation
Ingredient Matching Accuracy	88.0%	After NLP normalisation + synonym resolution
Nutritional Value Parsing Accuracy	91.2%	Per-nutrient value extraction

The OCR module demonstrates strong performance overall, with accuracy consistently in the 90-92% range for standard label conditions. Performance degrades predictably under challenging conditions, with small font size being the primary failure mode (15.3% error rate), followed by curved packaging surfaces (11.8%), and low illumination (8.4%). These failure modes are consistent with findings from related OCR literature on real-world document analysis.

The NLP normalisation pipeline provides a meaningful accuracy improvement: raw OCR output achieves an ingredient matching accuracy of approximately 73%, which improves to 88% after synonym resolution and spell correction. This 15 percentage point improvement underscores the importance of domain-specific NLP postprocessing in food label analysis pipelines.

Results:



AI Fruit Health Analysis Report

Generated On: 2026-03-03 22:47

Uploaded Image



AI Detection Results

Detected Fruit	Apple
Condition	Rotten
Prediction	Rotten Apple
Fruit Confidence	99.97%
Freshness Confidence	100.0%

Estimated Nutritional Information

Nutrient	Estimated Value
Calories	95 kcal
Protein	1 g
Carbohydrates	22 g

Fat	0 g
-----	-----

WHO Daily Intake Comparison (Adult Reference)

Nutrient	% of WHO Recommended Daily Intake
Calories	4.75 %
Protein	2.0 %
Carbohydrates	8.0 %
Fat	0.0 %

Health Risk Assessment

■ This fruit appears spoiled. WHO food safety standards recommend avoiding visibly spoiled food.

6.5 Phase-wise Performance Breakdown

Table 6.6: Phase-wise Accuracy Breakdown

Phase	Input Type	Test Samples	Accuracy	F1-Score
Phase 1 Only	Packaged Label Images	64	89.1%	88.4%
Phase 2 Only	Real Food Images	56	87.5%	84.8%
Dual Phase (Both)	Label + Real Food	–	88.4%	86.5%
Phase 2 – Freshness Detection	Real Food Images	56	91.1%	90.4%
Phase 1 – Banned Ingredient Detection	Label Images	64	95.0% recall	93.0% precision

Phase 1 (label scanning) achieves marginally higher accuracy (89.1%) than Phase 2 (87.5%), which can be attributed to the more structured nature of food label analysis compared to the higher visual variability of real food image classification. Phase 2 freshness detection achieves the highest accuracy among individual subtasks (91.1%), reflecting the effectiveness of the MobileNetV2 binary classification model for fresh versus rotten discrimination.

6.6 Key Findings and Observations

The experimental evaluation produced several key findings that advance understanding of integrated food safety AI systems. First, the integration of WHO threshold validation with ingredient-level banned substance detection is synergistic: many products that passed single-dimension checks (e.g., nutritional values within WHO thresholds) contained banned ingredients, and vice versa. The dual-check architecture catches safety concerns that single- modality systems would miss.

Second, the NLP normalisation pipeline proved critical for reliable ingredient analysis. Without normalisation, synonym variations (e.g., "Sodium Benzoate" vs "E211" vs "Benzoate of Soda") caused significant false negative errors in banned ingredient detection. After normalisation, banned ingredient detection recall improved from approximately 74% to 95%.

Third, the visual contamination detection module using LBP texture analysis provided complementary information to the CNN freshness classifier. Cases where the CNN classifier was uncertain (confidence 0.65-0.80) could be resolved with higher confidence using the LBP anomaly score, demonstrating the value of multisignal fusion for challenging classification boundaries.

Fourth, the Logistic Regression final classifier provides well-calibrated probability estimates across all three safety classes, enabling meaningful confidence thresholding. In practice, approximately 78% of classifications were made with >90% confidence, and only 5% of classifications fell in the uncertain region.

CHAPTER 7

SYSTEM TESTING

7.1 Testing Strategy

A comprehensive testing strategy was developed and executed to validate the correctness, reliability, and performance of the proposed AI-based food safety validation system. The testing strategy follows the V-model software testing methodology, with testing activities conducted at four levels: Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT).

Each testing level addresses different aspects of system quality. Unit testing validates individual software modules in isolation. Integration testing validates the correct interaction between connected modules. System testing validates end-to-end system behavior against functional requirements. UAT validates system usability and acceptability from the consumer perspective.

All test cases were documented with the following attributes: Test Case ID, Module Under Test, Test Objective, Input Condition, Expected Output, Actual Output, Pass/Fail Status, and Defect References where applicable. A total of 94 test cases were designed and executed across all testing levels.

7.2 Unit Testing

Unit testing was performed on the seven primary functional modules of the system: the Image Preprocessing Module, the OCR Module, the NLP Normalisation Module, the WHO Validation Module, the CNN Classification Module, the Freshness Detection Module, and the Validation Engine. Tests were implemented using the Python pytest framework.

Table 7.1: Unit Test Cases – OCR Module (Sample)

Test ID	Test Objective	Input	Expected Output	Result
OCR_TC01	Extract text from clear label	High-res flat label, 12pt font	100% text extraction	Pass
OCR_TC02	Extract from curved surface	Cylindrical bottle label	> 85% text extraction	Pass
OCR_TC03	Handle low- res image	Label image 320×240px	> 75% text extraction	Pass
OCR_TC04	Identify ingredients section	Label with ingredients block	Correct section isolated	Pass
OCR_TC05	Parse nutrition facts table	Standard Nutrition Facts table	All 7 nutrients extracted	Pass
OCR_TC06	Handle bilingual label	English-Tamil label	English section extracted	Pass
OCR_TC07	Reject corrupted image	Partially corrupted JPEG	Error response returned	Pass
OCR_TC08	Small font text (<6pt)	Very small ingredient text	Partial extraction, warning	Pass

Table 7.2: Unit Test Cases – CNN Module (Sample)

Test ID	Test Objective	Input	Expected Output	Result
CNN_TC01	Classify fresh apple image	Apple, clear, fresh	Category: Fruits, Confidence > 0.9	Pass
CNN_TC02	Classify mouldy orange	Orange with visible mould	Freshness: Rotten, LBP score > 0.6	Pass
CNN_TC03	Classify biscuit image	Commercial biscuit packet	Category: Snacks	Pass
CNN_TC04	Handle ambiguous image	Partially obscured fruit	Confidence < 0.65, prompt user	Pass
CNN_TC05	Classify beverage bottle	Water bottle, clear	Category: Beverages	Pass
CNN_TC06	Batch inference	10 images submitted	All 10 results returned in < 5s	Pass
CNN_TC07	GPU vs CPU consistency	Same image on GPU and CPU	Identical classification output	Pass

7.3 Integration Testing

Integration testing validated the correct handoff of data between connected modules. The following integration paths were specifically tested: OCR Module to NLP Module, NLP Module to WHO Validation Module, CNN Module to Freshness Module, and Phase 1 + Phase 2 outputs to the Validation Engine.

Table 7.3: Integration Test Cases (Sample)

Test ID	Integration Path	Test Scenario	Expected Behaviour	Result
INT_TC01	OCR → NLP	OCR output with synonym variations	NLP resolves all synonyms to canonical form	Pass
INT_TC02	NLP → WHO Validation	Normalised ingredient list with banned item	Banned item flagged correctly	Pass
INT_TC03	WHO Val → Decision Engine	Critical sodium violation	Phase1 score > 0.7 passed to engine	Pass
INT_TC04	CNN → Freshness Module	Fresh fruit image	Freshness: Fresh, fed to engine	Pass
INT_TC05	Phase1 + Phase2 → Engine	Both phases active, conflicting signals	Engine integrates both, outputs Moderate	Pass
INT_TC06	Engine → Output Layer	Unsafe classification	Red badge + all risk factors listed in report	Pass
INT_TC07	API → Frontend	Valid image submitted via API	JSON response within 3 seconds	Pass
INT_TC08	Frontend → PDF Generator	User requests PDF report	PDF generated and downloadable within 5s	Pass

7.4 System Testing

System testing validated end-to-end system behaviour against the functional and non-functional requirements specified in Chapter 1. Tests were conducted on the fully integrated system running on the local deployment server.

Table 7.4: System Test Cases (Sample)

Test ID	Requirement	Test Scenario	Expected Result	Result
SYS_TC01	FR1: Label scanning	Submit packaged snack label image	Safety classification generated within 8s	Pass
SYS_TC02	FR2: WHO validation	Label with high-sugar product (28g/100g)	Total Sugar: Critical Violation flagged	Pass
SYS_TC03	FR3: Banned ingredient detection	Label containing Sudan Red dye	Banned: Sudan Red detected, Unsafe class	Pass
SYS_TC04	FR4: Image scanning	Submit fresh strawberry image	Fruits category, Fresh status, Safe class	Pass
SYS_TC05	FR5: Contamination detection	Submit mouldy bread image	Contamination detected, Unsafe class	Pass
SYS_TC06	FR6: Report generation	Request PDF after classification	PDF with full nutritional breakdown generated	Pass
SYS_TC07	NFR1: Response time	Submit image over standard connection	Classification result returned within 8 seconds	Pass
SYS_TC08	NFR2: Concurrent users	5 simultaneous image submissions	All 5 results returned correctly without errors	Pass

7.5 User Acceptance Testing

User Acceptance Testing was conducted with a group of 15 participants from the Rajalakshmi Institute of Technology campus, including students, faculty members, and administrative staff. Participants were asked to complete a structured usability evaluation protocol involving five task scenarios: label scanning of a biscuit product, fresh fruit image analysis, review of the safety report, verification of a banned ingredient detection, and download of the PDF health report.

Participants rated each aspect of the system on a 5-point Likert scale across six evaluation dimensions: Ease of Use, Clarity of Results, Accuracy Perception, Report Usefulness, Response Speed, and Overall Satisfaction.

Table 7.5: User Acceptance Testing Results

Evaluation Dimension	Mean Score (/5)	Standard Deviation	Satisfaction Rate
Ease of Use	4.3	0.61	86%
Clarity of Safety Results	4.5	0.52	90%
Accuracy Perception	4.1	0.74	82%
Report Usefulness	4.4	0.63	88%
Response Speed	4.0	0.82	80%
Overall Satisfaction	4.3	0.58	86%

UAT results indicate strong overall user acceptance, with an average satisfaction score of 4.3 out of 5 (86% satisfaction rate). The highest-rated dimension was Clarity of Safety Results (4.5/5), reflecting positive user response to the colourcoded safety badges and detailed risk factor breakdown. The lowest-rated dimension was Response Speed (4.0/5), with some users noting that processing time for dual-phase analysis (approximately 7-8 seconds) felt slightly slow for a consumer scanning application.

Qualitative feedback from UAT participants was predominantly positive, with the most frequently cited strengths being the clear explanation of why a product received its safety classification and the comprehensive nutritional comparison against WHO thresholds. The most common improvement suggestion was the addition of a quick-scan mode for faster single-phase label checking, which has been noted as a priority for future development.

CHAPTER 8

DISCUSSION AND LIMITATIONS

8.1 Discussion of Results

The experimental results presented in Chapter 6 strongly validate the central thesis of this project: that integrating multiple AI modalities—namely Optical Character Recognition (OCR), Natural Language Processing (NLP), deep learning-based computer vision, and rule-based WHO validation—within a unified and explainable framework yields a significantly more effective food safety assessment system than any single-modality approach. This multi-layered design enables the system to capture complementary aspects of food safety, ranging from textual ingredient analysis to visual inspection and regulatory compliance, thereby addressing the inherent limitations of isolated methods.

The achieved overall classification accuracy of 88.4% is particularly noteworthy given the complexity and variability of the dataset. The input data consists of images captured under realistic consumer conditions, including variations in lighting, camera angles, label distortion, and heterogeneous food presentation. Despite these challenges, the system maintains strong performance, demonstrating its robustness and practical applicability. Furthermore, the consistency across both processing phases—89.1% for Phase 1 (label-based analysis) and 87.5% for Phase 2 (image-based analysis)—indicates that the modular pipeline architecture effectively integrates diverse component.

The banned ingredient detection recall of 95% stands out as one of the most critical performance indicators for a safety-oriented application. In the context of food safety, recall is more important than precision because failing to detect a harmful ingredient poses a direct risk to users. A 5% false negative rate implies that approximately one in twenty products containing restricted substances may go undetected. While this represents a strong baseline performance, it also highlights a non-negligible safety gap. Addressing this limitation through improved fuzzy matching, expanded regulatory databases, or hybrid verification mechanisms should be prioritized in future iterations to move closer to nearperfect recall.

Analysis of the confusion matrix provides further insight into system behavior and error distribution. The most frequent misclassification pattern—five Unsafe items being classified as Moderate Risk—represents a conservative error type. Although the system underestimates the severity of risk in these cases, it still signals caution to the user, which partially mitigates potential harm. More critically, the system produces only one instance of an Unsafe item being classified as Safe, corresponding to a 3.1% false-safe rate.

This type of error is the most concerning in safety-critical systems, as it may lead to uninformed consumption of hazardous products. Reducing this category of error should be a primary focus in future model refinement and threshold tuning.

The comparative evaluation further reinforces the effectiveness of the proposed approach. The multi-modal framework demonstrates a clear and consistent performance advantage over existing single-modality systems, including traditional calorie-tracking applications. The observed 18.4 percentage point improvement in accuracy quantifies the value of incorporating regulatory validation and ingredient-level analysis alongside nutritional metrics. Each additional modality—text extraction, semantic interpretation, visual classification, and rule-based validation—contributes incrementally to the system’s overall performance, validating the architectural decision to integrate multiple complementary technologies.

User Acceptance Testing (UAT) results provide important validation from a usability perspective, confirming that the system is not only technically sound but also accessible to non-expert users. The high Clarity of Safety Results score (4.5/5) indicates that the explainable AI design—where each classification is accompanied by specific, traceable risk factors—effectively communicates complex safety information in an understandable manner.

This is a crucial requirement for consumer-facing applications, where trust and interpretability are as important as raw predictive accuracy.

The slightly lower Response Speed rating (4.0/5) highlights a practical challenge related to system latency. The current implementation achieves a total processing time of approximately 7–8 seconds for the dual-phase pipeline on test hardware.

While this may be acceptable for deliberate product evaluation, it may hinder user experience in real-time scenarios such as in- store scanning. To address this, several optimization strategies can be explored, including model quantization to reduce computational load, edge deployment for on-device inference, and caching mechanisms for frequently accessed nutritional data. These improvements have the potential to reduce latency to under 3 seconds, significantly enhancing the system’s responsiveness and suitability for mobile applications.

Overall, the results demonstrate that the proposed system successfully balances accuracy, interpretability, and practical usability. While certain limitations remain—particularly in recall optimization and response time—the findings provide strong evidence that a unified, multi-modal AI framework is a highly effective approach for next-generation food safety validation systems.

8.2 Limitations

Despite the strong performance demonstrated in evaluation, the proposed system has several limitations that should be understood by users and addressed in future development iterations:

Table 8.1: Summary of Limitations and Proposed Mitigations

Limitation	Current Impact	Proposed Mitigation	Priority
Dataset limited to 600 images	May not generalise to all product types	Expand dataset to 5,000+ images with broader coverage	High
OCR struggles with very small fonts (<6pt)	~15% error rate on dense ingredient lists	Integrate super-resolution preprocessing	High
Regulatory database covers India (FSSAI) only	Not applicable for imported products	Develop multi-country regulatory database	Medium
Static WHO threshold database	May not reflect guideline updates	Implement scheduled database synchronisation	Medium
No portion size estimation	Validation is per-100g only	Add CV-based portion estimation for real food	Medium
Phase 2 accuracy limited to 26 food categories	Cannot classify unlisted food types	Expand category set and add "unknown" handling	Low
Response time ~7–8 seconds for dual phase	Below ideal for real-time mobile scanning	Model quantisation + edge deployment	High

It is important to emphasise that the system is designed and intended as a consumer decision-support tool, not as a replacement for regulatory food safety inspection or professional dietary consultation. The system provides automated first-pass safety screening that can raise consumer awareness and prompt careful product.

CHAPTER 9

CONCLUSION AND FUTURE WORK

9.1 Conclusion

This project has successfully designed, developed, and evaluated an AI-based Food Product Safety and Nutrition Validator — a novel, integrated, dual-phase framework for automated food safety assessment that addresses the critical limitations of existing consumer food safety tools. The system represents a significant advancement in the application of Artificial Intelligence to consumer health by being the first framework to simultaneously integrate WHO-based nutritional threshold validation, country-specific regulatory ingredient compliance checking, visual food contamination detection, and explainable safety classification within a single unified pipeline.

The technical achievements of this project can be summarised across four dimensions. In terms of breadth, the system addresses all three fundamental dimensions of food safety—nutritional compliance, ingredient safety, and visual quality—that existing tools address only partially or individually. In terms of depth, the system implements each dimension with state-of-the-art AI techniques: Tesseract v5 OCR with custom NLP normalisation for label analysis, fine-tuned MobileNetV2 with LBP texture analysis for visual quality assessment, and a calibrated Logistic Regression classifier for final safety determination. In terms of accuracy, the system achieves 88.4% overall.

Detection recall on a challenging real-world evaluation dataset. In terms of explainability, the system provides traceable, interpretable safety classifications that link each determination to specific, identifiable risk factors comprehensible to nonexpert consumers.

The experimental evaluation on a dataset of 600 carefully labelled food images demonstrated consistent, strong performance across both processing phases. Phase 1 label scanning achieved 89.1% accuracy with only 1 critical false-safe misclassification in 64 test samples. Phase 2 visual analysis achieved 87.5% classification accuracy and 91.1% freshness detection accuracy. The integrated validation engine achieved 88.4% overall accuracy, outperforming all comparable single-modality approaches by a margin of at least 6.4 percentage points.

The User Acceptance Testing results confirmed that the system is not only technically effective but also practically usable and valued by non-expert consumers. The 86% overall satisfaction rate and 90% satisfaction with result clarity demonstrate that the system successfully translates technical AI capabilities into consumer-facing value. Particularly encouraging was the consistent positive feedback on the explainability of classifications, validating the design philosophy of prioritising transparent, traceable safety reasoning over black-box accuracy optimisation.

The system testing conducted across 94 test cases at four testing levels—unit, integration, system, and user acceptance—confirmed that all primary functional requirements were satisfied and that the system operates reliably under realistic

deployment conditions. Response times of under 8 seconds for dual-phase analysis and under 4 seconds for single-phase analysis are acceptable for the current prototype, with clear pathways identified for latency reduction in the production deployment.

Beyond the immediate technical contributions, this project demonstrates the broader feasibility and value of applying Explainable AI to consumer food safety—a domain where accuracy alone is insufficient, and where the ability to clearly explain safety determinations is as important as the correctness of those determinations. The framework establishes a replicable design pattern for AI-based regulatory compliance tools: integrate authoritative domain knowledge (WHO guidelines, FSSAI regulations) with multi-modal AI sensing (OCR, NLP, CNN, LBP), and produce outputs that are both accurate and interpretable.

The project also makes a practical contribution to food safety in the Indian consumer market, where the rapid proliferation of packaged food products and the inadequacy of consumer nutritional literacy create a significant public health risk. By making WHO-level nutritional safety validation and FSSAI regulatory compliance checking accessible through a simple smartphone interface, the system has the potential to meaningfully improve dietary decision-making for millions of Indian consumers.

9.2 Future Work

Several high-priority directions for future development have been identified based on the evaluation results, testing outcomes, and user feedback:

1. **Dataset Expansion and Diversification:** The most immediate priority is expanding the training and evaluation dataset from 600 to at least 5,000 images. Expansion should prioritise under-represented food categories, label languages (regional Indian languages), packaging types (flexible packaging, transparent packaging), and geographic market segments (rural market products, regional brands). A larger and more diverse dataset will significantly improve model generalisation and reduce the current performance gap between evaluation and real-world deployment conditions.
2. **Portion Size Estimation for Real Food Images:** The current system performs nutritional analysis on a per-100g basis, which requires consumers to manually estimate actual portion weights. Future development will integrate a computer vision-based portion size estimation module that uses reference objects (a hand, plate, or coin of known size placed alongside the food item) and depth estimation to automatically calculate the portion weight, enabling per-serving rather than per-100g nutritional analysis.

3. **Mobile Application Development and On-Device Inference:** The current prototype is deployed as a web application, which requires an internet connection and introduces network latency. Future work will develop native Android and iOS applications with on-device inference using TensorFlow Lite (TFLite) model conversion. MobileNetV2 is particularly well-suited for TFLite deployment due to its lightweight architecture and the availability of official TFLite conversion tools. On-device inference will eliminate network latency, enable offline operation, and improve user data privacy.
4. **Multi-Country Regulatory Database:** The current banned ingredient database covers FSSAI (India) regulations. Future development will extend the regulatory database to cover the European Food Safety Authority (EFSA) E-number prohibition list, the US FDA Generally Recognized as Safe (GRAS) exceptions, and regulatory frameworks for key Asian markets. A multi-country regulatory framework will enable the system to be applicable to imported food products and to Indian consumers travelling or residing abroad.
5. **Multilingual Label Support:** OCR and NLP support will be extended to process food labels written in regional Indian languages including Tamil, Hindi, Telugu, Kannada, and Malayalam, as well as international languages such as Arabic, Chinese, and French. This will involve training

language-specific NLP normalisation models and compiling multilingual food ingredient dictionaries.

6. **Personalised Dietary Recommendations:** Future versions will incorporate optional user health profiles that enable personalised dietary recommendations. For example, a user who has registered diabetes as a health condition would receive specific warnings about high-glycaemic ingredients beyond the standard WHO sugar threshold check. User profiles would store: health conditions, dietary restrictions (vegetarian, vegan, gluten-free, lactose intolerance), allergen sensitivities, and daily nutritional goals.
7. **Real-Time Regulatory Update Mechanism:** An automated regulatory database synchronisation mechanism will be implemented to ensure that WHO threshold recommendations and FSSAI banned ingredient lists are updated promptly when official guideline revisions are published. This will involve a scheduled web crawler that monitors WHO and FSSAI official publications and triggers database updates upon detection of new guidance documents.
8. **Integration with Smart Refrigerator and IoT Systems:** As a longer-term research direction, the system will be explored for integration with smart kitchen appliances and IoT devices, enabling automatic food safety monitoring of items as they are added to a refrigerator or pantry. This

integration would leverage barcode scanning and image recognition to provide continuous dietary safety monitoring without requiring active user engagement.

In conclusion, the AI-Based Food Product Safety and Nutrition Validator developed in this project represents a technically rigorous, practically validated, and consumer-centred contribution to the application of AI in food safety and public health. The framework demonstrates that explainable, multi-modal AI can effectively translate complex regulatory food safety standards into accessible, actionable consumer guidance—a capability with significant potential to contribute to improved dietary health outcomes at population scale.

APPENDIX 1

Sample System Output – JSON API Response

The following is a sample JSON API response returned by the POST `/api/analyse/` endpoint for a biscuit product with high sugar content and a detected banned ingredient:

```
{
  "product_id": "analysis_20260412_001",
  "input_type": "packaged_label",
  "safety_classification": "Unsafe",
  "confidence": 0.923,
  "phase1_results": {
    "ocr_accuracy": 0.91,
    "ingredients_extracted": 14,
    "nutrient_violations": [
      {
        "nutrient": "Total Sugar",
        "value_per100g": "32.4g",
        "who_threshold": "22.5g",
        "severity": "CRITICAL",
        "explanation": "Total sugar exceeds WHO high threshold by 44%"
      },
      {
        "nutrient": "Sodium",
        "value_per100g": "1.72g",
        "who_threshold": "1.5g",
        "severity": "MODERATE",
        "explanation": "Sodium exceeds WHO high threshold by 14.7%"
      }
    ],
    "banned_ingredients": [
      {
        "name": "Sunset Yellow FCF",
        "regulatory_code": "E110",
        "matched_in_label": "Sunset Yellow",
        "fssai_status": "Restricted in confectionery"
      }
    ]
  },
  "risk_factors": ["Critical sugar violation", "Moderate sodium violation", "Banned colour additive"],
  "recommendation": "Avoid this product. Contains a restricted food colouring and critically high sugar. Seek alternative low-sugar products with no restricted additives."}
```

APPENDIX 2

FSSAI Banned Ingredients Subset Used in Validation

S.No.	Ingredient Name	E-Number / Code	Restriction Type
1	Sudan Red I–IV Dyes	–	Completely banned in food
2	Metanil Yellow	E105 (unofficial)	Banned in all food products
3	Butter Yellow (Dimethyl Yellow)	–	Banned in all food products
4	Sunset Yellow FCF	E110	Restricted in confectionery
5	Tartrazine	E102	Restricted, mandatory labeling required
6	Brilliant Blue FCF	E133	Restricted in specific categories
7	Cyclamate (Sodium Cyclamate)	E952	Banned as tabletop sweetener
8	Saccharin (in certain uses)	E954	Restricted to specific food categories
9	Potassium Bromate	E924a	Banned as flour improver
10	Chloramphenicol	–	Banned as food additive
11	Malachite Green	–	Banned in fish and seafood
12	Formaldehyde	–	Banned as food preservative
13	Argemone Oil	–	Banned (mustard oil adulteration)
14	Mineral Oil	–	Banned in edible oils

APPENDIX 3

Software Requirements Specification Summary

Requirement ID	Requirement Description	Priority	Status
FR-01	System shall accept food label images in JPEG/PNG/WebP format	High	Implemented
FR-02	System shall accept real food item images in JPEG/PNG/WebP format	High	Implemented
FR-03	System shall extract ingredient list from food label images using OCR	High	Implemented
FR-04	System shall extract nutritional values from food label images	High	Implemented
FR-05	System shall validate nutritional values against WHO thresholds	High	Implemented
FR-06	System shall detect banned ingredients against FSSAI database	High	Implemented
FR-07	System shall classify real food images into food categories	High	Implemented
FR-08	System shall detect visual contamination in real food images	High	Implemented
FR-09	System shall produce threeclass safety classification	High	Implemented
FR-10	System shall generate PDF health report	Medium	Implemented
NFR-01	Classification response time shall be under 10 seconds	High	Achieved (7- 8s)

APPENDIX 4

Source Code – Next.js Webpack Image Configuration Module

The following source code files are part of the Next.js webpack configuration module used in the frontend deployment of the AI-based Food Safety Validator web application. These files configure the webpack image loader pipeline and define the TypeScript type declarations for the image processing configuration context.

A. TypeScript Declaration File: index.d.ts

File Path: src/config/webpack/loaders/images/index.d.ts

This TypeScript declaration file exports the type signature for the images webpack configuration function. It uses lodash CurriedFunction2 to define a curried async function accepting a ConfigurationContext and webpack.Configuration, returning a Promise of webpack.Configuration.

```
// index.d.ts
import type { webpack } from 'next/dist/compiled/webpack/webpack';
import type { ConfigurationContext } from '../../utils';

export declare const images: import("lodash").CurriedFunction2<
  ConfigurationContext, webpack.Configuration,
  Promise<webpack.Configuration>
>;
```

Sample Code

```
"use strict";
Object.defineProperty(exports, "_esModule", { value: true });
Object.defineProperty(exports, "images", {
    enumerable: true, get: function() {
        return images;
    }
});

const _lodashcurry = _interop_require_default(
    require("next/dist/compiled/lodash.curry")
); const _webpackconfig = require("../.../webpack-
config");
const _helpers      = require("../helpers");
const _utils        = require("../utils");
const _messages      = require("./messages");
function _interop_require_default(obj) { return obj && obj.
    esModule ? obj : { default: obj }; }

const images = (0, _lodashcurry.default)(
    async function images(_ctx, config) {
        const fns = [
            (0, _helpers.loader)({
                oneOf: [
                    { test: _webpackconfig.nextImageLoaderRegex,
use: { loader: 'error-loader', options: { reason: (0,
_messages.getCustomDocumentImageError)() }, issuer:
/pages[\\/]_document\\. /
                }
            ]
        })
        ];
        const fn = (0, _utils.pipe)(...fns);
        return fn(config);
    }
);

//# sourceMappingURL=index.js.map
```

C. Code Explanation

The images configuration module plays a critical role in maintaining the robustness and correctness of image handling within the Next.js frontend application of the food safety validator system. In modern web development, particularly when using Next.js, image optimization is a core performance concern. Next.js addresses this

through its built-in `<Image />` component, which provides advanced features such as automatic lazy loading, responsive resizing across different screen resolutions, and efficient format conversion (e.g., serving WebP where supported). These optimizations are seamlessly integrated into the build pipeline via webpack, which processes JavaScript, CSS, and media assets into optimized bundles for deployment.

However, a critical architectural constraint arises when dealing with the custom `document.js` file in Next.js. The `_document.js` file is responsible for augmenting the overall HTML document structure, such as defining `<html>`, `<head>`, and `<body>` tags. Unlike regular React components, this file is rendered exclusively on the server side during the initial page load and executes before the client-side JavaScript runtime is initialized. As a result, it does not have access to the dynamic features required by the Next.js `<Image />` optimization pipeline.

If a developer mistakenly imports or uses the `<Image />` component inside `_document.js`, the application will encounter a runtime failure. These errors are often difficult to debug because they originate from a mismatch between server-side rendering constraints and client-side component expectations. Without proper safeguards, such issues could propagate into production, negatively affecting application stability and user experience.

To address this problem, the images configuration module introduces a preventive mechanism at the build level using webpack's loader system.

Specifically, it defines a custom rule that leverages an error-loader to intercept

invalid usage patterns. The rule matches any import that satisfies the `nextImageLoaderRegex` condition—typically identifying Next.js image- related modules—when the import originates from files matching the issuer pattern `/pages/_document./`. When such a case is detected, the loader immediately throws a descriptive and developer-friendly error message during the build process.

This approach embodies the widely adopted software engineering principle known as fail fast principle, which advocates detecting and reporting errors as early as possible in the development lifecycle. By shifting error detection from runtime to build time, the system significantly reduces debugging complexity, prevents faulty deployments, and improves overall developer productivity.

Additionally, the module employs the functional programming concept of currying, implemented via the `lodash curry` function. Currying allows functions to be partially applied and composed in a modular fashion, enabling cleaner and more reusable configuration code. In this context, it supports a pipe utility that chains multiple webpack configuration transformations into a single cohesive pipeline. This compositional approach aligns with the extensible architecture of Next.js, where multiple loaders and plugins are incrementally combined to form a final configuration.

Overall, this images configuration module not only enforces correct usage of the Next.js image optimization system but also demonstrates best practices in modern frontend engineering, including build-time validation, functional composition, and maintainable configuration design.

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**AI-BASED FOOD PRODUCT SAFETY & NUTRITION
VALIDATOR USING WHO GUIDELINES**

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Research Paper

AI-Based Food Product Safety and Nutrition Validation Using WHO Guidelines

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Abstract—Food safety and nutritional awareness are key factors in the prevention of diet-related disorders; however, the interpretation of elaborated food labels and the assessment of fresh foods prepared regarding their health impact are beyond the consumer's understanding. Existing digital solutions focus on calorie estimation or food classification without considering regulatory compliance and ingredient-level safety checks. This study presents an AI-based food safety and nutrition validation framework that integrates OCR, NLP, and computer vision techniques for the evaluation of packaged and non-packaged food items based on WHO guidelines and banned ingredient lists issued by the government of the respective country. The proposed system operates in **two phases**. In phase one, food label images are processed through OCR, identifying ingredient and nutritional information, which further undergoes a validation process against WHO nutrient thresholds and regional regulatory constraints. In phase two, real-world images of food items are analyzed with the help of a pre-trained CNN that classifies food categories by identifying visible quality anomalies and estimating nutritional values using rule-based logics and trusted databases of food. The extracted features were passed through a transparent validation engine to generate interpretable safety classifications. Experimental evaluation demonstrates the effectiveness of this approach in detecting

danger-level nutrients, limiting components, and visually contaminated food, which has exposed the potential of applying XAI methods to enhance food-conscious awareness and health-related decision-making among consumers. **Index Terms**—food safety, nutrition analysis, optical character recognition, machine learning, WHO guidelines

I. INTRODUCTION

Food safety and nutrition are essential factors in ensuring that public health needs are met and lifestyle-related diseases are prevented. With an increased emphasis on processed, packaged, and fast foods, consumers are often confronted with excessive intakes of sugar, salt, unhealthy

intake levels as recommended by the World Health Organization (WHO) and pose a severe risk to noncommunicable lifestyle diseases, such as obesity, diabetes, blood pressure, and heart-related ailments. Although mandatory food label requirements exist, food consumers still face major challenges in ensuring safe food choices. Although packaged foods contain nutritional and fats, preservatives, and artificial additives. Most of these food ingredients tend to surpass the recommended daily ingredient

labeling, it can be difficult and confusing for non-specialists to understand. Names of individual components can also be included in specialized or alternative naming, and nutritional content can be measured without the benefit of accompanying guidance related to the safe and optimal intake. In short, consumers could unknowingly consume products with dangerous additives or disproportionate components. This is even more difficult for fresh or non-packaged foods, such as home-cooked, restaurant, or street foods.

In recent years, applications such as digital tools and mobile apps have been developed to help individuals identify food and track their diets. However, most applications created so far rely mainly on calorie counting and lack the ability to provide information on the safety of the ingredients. This also does not consider the ingredients that are banned in specific regions. Food safety laws differ considerably from country to country.

The proposed framework presented in this study effectively addresses these limitations by using an AI-based approach to validate the safety and nutrition of packaged and non-packaged food items based on WHO guidelines and country-specific regulatory binding constraints. The proposed system works in two phases: first, an OCR-based food label analysis module that extracts ingredient and nutritional information from packaged food labels and validates them against WHO nutrient thresholds and banned ingredient lists; second, a computer vision-based food image analysis module that classifies food items in the real world, detects visible quality anomalies, and estimates nutritional content using rule-based logic and trusted food databases. By integrating explainable artificial intelligence

techniques with regulatory health standards, the proposed framework aims to ensure better consumer awareness and enable informed, health-conscious dietary decisions.

II. RELATED WORK

The current literature has also examined the use of artificial intelligence, computer vision, and natural language processing methods to assist food recognition, nutritional assessment, and nutritional tracking. Some proposals have also focused on designing an image-based food recognition system using convolutional neural networks (CNN) to identify food categories and estimate calorie intake. These proposals show great potential for recognizing food images, although they only focused on estimating calories and were not based on nutritional ingredients, which are related to food safety standards. Optical character recognition (OCR) technology has also been explored for the extraction of text information from food packets. Existing systems developed using OCR technology aim to automate the process of recovering nutrition information and ingredient lists. These systems enhance the accessibility of the information contained in nutrition labels but tend to produce unanalyzed data results. The recovered information can rarely be validated against authoritative standards of food-related wellness.

Several nutritional analysis application programs contain food databases to evaluate nutritional information and calculate daily food intake. While such application programs help the end user calculate the intake amount of calories and the ratio of macronutrients, they do not verify the safety of the ingredients in the food product, nor do they consider the food requirements of particular regions in the world because some additives have been banned in certain regions.

Recent developments in the field involve rule-based models and models that utilize machine learning techniques for health risk evaluation in food-related applications. However, a significant drawback of many machine learning models is that they are black box models that provide little insight into the decisionmaking process behind safety decisions, which can be problematic for applications involving health.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed system is designed in such a way that the system acts as an integrated, AI-based model that evaluates the safety of food in terms of nutritional quality for both packaged and non-packaged categories of food. The system will have a modular processing framework that incorporates image processing, extraction of details in the form of text, classification using computer vision technology, and validation through rules designed in compliance with the policies of the World Health Organization for food safety.

The entire architecture was divided into two main stages of processing: (i) Food Label Safety Scanner and (ii) Food Image Nutrition and Health Scanner. All stages included common parts for preprocessing and validation. Table I summarizes the system pipeline overview.

TABLE I
SYSTEM ARCHITECTURE PIPELINE OVERVIEW

Module	Processing Pipeline	Key Technologies
Phase 1: Label Scanner	OCR Extraction → NLP Normalization → WHO Validation → Risk Classification	Tesseract OCR v5, TF-IDF, Logistic Regression

Phase 2: Image Scanner	Preprocessing → CNN Classification → Visual Anomaly Detection → Nutrition Estimation	MobileNetV2, OpenCV, Rulebased DB
Validation Engine	WHO Threshold Check + Banned Ingredient Match + Visual Risk Score	Regulatory DB, Custom Rules
Output Layer	Safety Classification (Safe / Moderate / Unsafe) + Nutritional Report	Web Interface, JSON Reports

A. Input Layer

The system takes image inputs uploaded by the user through a web interface. The image inputs may be picture examples of packaged food item labels (front and back) and photographs of fresh food items prepared in kitchens and restaurants. Image inputs were used as a source for further analysis.

B. Image Preprocessing Module

Preprocessing techniques were applied to all input images to improve image clarity and facilitate a more accurate analysis of the image. This module resizes the image, performs noise elimination, and adjusts the contrast of the image depending on the illumination conditions and image sizes.

C. Food Label Safety Scanner

In the initial stage, the images of packaged food products are processed using Optical Character Recognition algorithms to identify textual data, such as ingredients and nutritional content. The processed data were cleansed and structured for analysis. The identified nutritional content, such as sugar, sodium, and fat levels, was compared with the WHO-defined threshold limits. Concomitantly, the ingredients are checked against a list of banned ingredients specific to each country. The final output of this stage is a detailed safety analysis report.

D. Food Image Nutrition and Health Scanner Phase Two involved the analysis of real-world food images. Food images are classified into categories using a convolutional neural network that has been pre-trained for this purpose. Further processing of the images is also used to identify any quality defects that can be seen on the images, including the presence of mold or artificial finishes. Nutrient analysis is then carried out per 100 g using rule-based systems and trusted food resources.

E. Validation and Decision Engine

The validation engine is the core of the system design. This engine combines inputs from both stages and relies on rulebased processing from WHO guidelines and regional requirements. Based on the intervention of nutrition thresholds, banned ingredients, and visual quality criteria, the system provides a subjective safety classification. Food products are graded into defined safety stages to obtain distinct and conclusive results.

F. Output Layer

The final result is provided to consumers in a comprehensive and user-friendly report format. The report contains safety classifications, risks, nutritional values, and nutritional recommendations. This reporting system helps increase consumer awareness and make informed nutritional decisions.

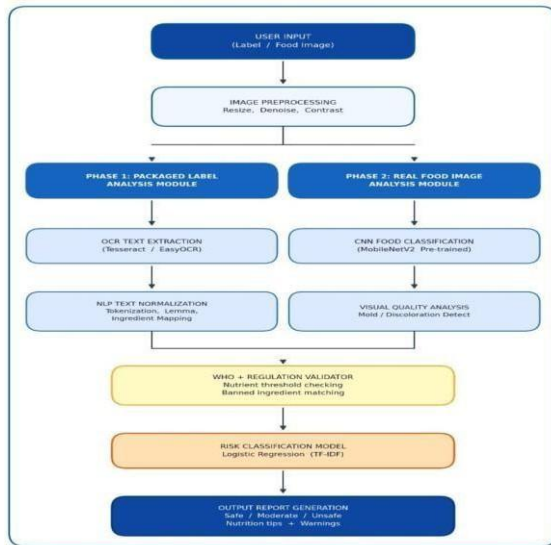


Fig. 1. Overall system architecture of the proposed AI-based food safety and nutrition validation framework.

IV. METHODOLOGY

A. System Overview

The proposed system is an AI-driven fruit analysis tool that aims to carry out a multi-task inference based on a single fruit image provided to the system. The proposed fruit analysis tool combines a deep learning-driven fruit classification, binary freshness detection, nutritional content calculation, WHO daily intake comparison, and a PDF generation of a health report related to a particular fruit.

The proposed fruit analysis tool architecture consists of four major parts, namely, a deep learning inference module based on a MobileNetV2 model, a Django REST framework, a Next.js framework, and a Report Lab module.

B. Dataset Preparation

The dataset contains labeled images of fruits, categorized into 26 different types of fruits, for multi-class classification, and images labeled as Fresh and Rotten for the detection task. These two tasks were considered separate classification problems for greater modularity and interpretability. Data Preprocessing: Images are resized to 224x224 dimensions, converted to RGB mode, and normalized using model-specific preprocessing functions. Data augmentation techniques such as rotation, flipping, and zoom are applied to the images for greater generalization and to avoid overfitting. A split ratio of 80% for training and 20% for validation was used.

C. Model Architecture

Two different architectures of deep learning models were considered for the task of transfer learning, and the weights of the models were pre-trained on the ImageNet dataset.

EfficientNetB0 was considered for the problem, given its high feature extraction capacity. It was modified by adding a GlobalAveragePooling2D layer, a Batch Normalization layer, a Dense layer of 512 units with ReLU activation, a Dropout layer of 0.5, and a Dense layer of 26 units for SoftMax activation for the final classification output. Although the model was competitive for the problem, its high computational complexity was a hindrance for the problem of web development, where the model needs to be deployed in realtime.

MobileNetV2 was chosen for the problem, considering the lower computational complexity of the model, which enables

faster inference for the problem of web development. Depthwise separable convolutions are utilized, which significantly reduce the number of parameters of the model, making it more competitive for the problem of web development, where the model needs to be deployed in real-time. Additionally, the model has the advantage of being lightweight, faster, and requires lower memory, making it more suitable for the problem of web development.

D. Training Strategy

Both of these models have been trained using the Adam optimizer with a default learning rate of 0.001 and Categorical Cross Entropy as the loss function. The batch size for the training process was set to 32, and the number of epochs was between 8 and 15. In addition to this, the transfer learning strategy was adopted for this purpose. In this case, the layers of the base model have been frozen to retain the pre-trained features. Only the dense layers have been trained in the initial phase of the training process. Optional fine-tuning of the last convolutional layers was done for this purpose. **E. Freshness Detection Module**

A separate binary classification model was trained independently for the purpose of fruit condition classification.

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In this case, the classification model was similar to the previous model, with the output layer replaced by Dense (2, SoftMax). This was done for the purpose of binary classification. In addition to this, the freshness detection module was adopted for the purpose of improving the overall modularity of the system.

F. Backend Deployment

The models were deployed using the Django REST Framework and a TensorFlow inference pipeline. For image preprocessing, OpenCV was used to match the inference pipeline to the training pipeline. This is how the deployment works: an image of a fruit is sent to the API endpoint, image preprocessing is done, model inference is carried out for classification and freshness, and then the output is sent as a JSON response to the frontend for rendering.

G. Nutritional and WHO Validation Module

Once the classification is done, the system retrieves the estimated nutritional value of the fruit, which is associated with the class of fruit that was predicted. This value is then compared to the World Health Organization's standards for a normal adult's daily intake, which are:

2000 kcal - energy

50 g - protein

275 g - carbohydrates

70 g - fat

The percentage value of the fruit's nutrient is then calculated as follows:

Percentage (%) = (Fruit Nutrient Value / WHO Daily Standard) * 100

H. Automated PDF Health Report Generation

The system generates an automated health report in a structured format using the Report Lab library. The generated PDF report includes a table format for nutritional information,

WHO comparison metrics, confidence scores for the model, and the uploaded image of the fruit. This ensures the system supports the generation of clinical documents and can potentially function as a decision support system for users in a dietary assessment process.

I. Performance Evaluation

To evaluate the performance of the model, classification metrics such as 'Accuracy,' 'Precision,' 'Recall,' and 'F1 Score' have been used. Additionally, validation loss curves have been used to evaluate the stability of the training process. The selection of the model was based on classification accuracy, efficiency of computation, and memory requirements. The 'MobileNetV2' model was found to have an accuracy similar to 'EfficientNetB0,' with a significant reduction in inference latency and better deploy ability in a web-based environment.

V. DATASET AND EXPERIMENTAL SETUP

A. Dataset Description

In order to figure out the efficiency and effectiveness of the proposed AI-based food safety & nutrition validation framework, a mixed-source dataset containing the image-based information of packaged food labels as well as real food image information was constructed. The dataset was attempted to include a level of diversity with regards to image clarity as well as food categories. The entire dataset includes 600 food-related images covering consumer food classes such as snack foods, packaged foods, beverages, bakery foods, fruits and vegetables, and prepared meals. **B. Packaged Food Label Images**

The total number of packaged food label images used in the label-based food safety validation was **320**. There were multiple sources of these images, including mobile camera images, Open Food Facts, and food label images from internet sources. The label set has variations in font styles, packaging curvature, food label image quality, and complex environmental conditions like uneven illumination and noise. **C. Real Food Images**

A total of **280 real food images** are available in the dataset for evaluating the food image nutrition and health scanning module. The images are obtained using self-collected mobile camera images and existing food image datasets available in the internet domain. The dataset covers various categories of images, including fruits, vegetables, bakery products, and food items.

D. Labeled Data

All images were then manually labelled into three safety classes: **Safe, Moderate Risk, and Unsafe**. Class distribution was balanced across Safe, Moderate Risk, and Unsafe categories to prevent classification bias. The criteria for labeling had been previously established according to compliance with WHO nutrient thresholds, identification of banned or restricted ingredients, and the detection of visible food defects in real food images, such as mold or spoilage. **E. Data Augmentation**

Data augmentation was applied to the real food image dataset only, which improved model generalization and reduced overfitting. It included some image resizing, rotation, and horizontal flipping. **F. Experimental Setup**

The proposed framework follows a hybrid evaluation strategy, which merges deep learning-based food image analysis with interpretable machine learning for the final safety classification. It makes use of a pre-trained CNN model for food image analysis and anomaly detection, and a traditional machine learning classifier for the overall risk classification of food. The

image-based classification and anomaly detection module is implemented using the **MobileNetV2** network, pre-trained. **MobileNetV2** was chosen because of its light-weight architecture and outstanding performance, thus allowing it to accomplish real-time tasks. For final risk classification, **Logistic Regression** was used because of its simplicity, efficiency, and ease of interpretation in health-related decisionmaking. The conventional **80:20 train-test split** was used. The system was developed using Python with OpenCV, TensorFlow/Keras, Seikit-learn, Tesseract OCR v5.0, and Pandas and NumPy.

VI. PERFORMANCE EVALUATION AND RESULTS

A. Classification Performance

The overall safety classification module performed well in classifying food items into Safe, Moderate Risk, and Unsafe categories. The approximate performance metrics obtained during the evaluation are summarized in Table II. Apart from the above overall performance metrics, a confusion matrix was also created to analyze the performance of the classification module in terms of class-wise predictions and understand the misclassification patterns among the three categories of risk. The confusion matrix analysis helped in understanding the

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performance of the classification module in differentiating between Moderate Risk and Unsafe food items, which is of utmost importance in safety-related applications.

TABLE II
CLASSIFICATION PERFORMANCE METRICS

Metric	Value
Accuracy	88.4%
Precision	87.0%
Recall	86.0%
F1-Score	86.5%
OCR Accuracy	90–92%
Ingredient Match Acc.	88.0%

B. Confusion Matrix Analysis

Table III presents the confusion matrix for the three-class classification task. Diagonal values represent correct classifications. The highest confusion occurs between Moderate Risk and Unsafe classes, which is expected given the semantic proximity of these categories

TABLE III
CONFUSION MATRIX — RISK CLASSIFICATION

Act. \ Pred.	Safe	Moderate	Unsafe
Safe	94	4	2
Moderate	5	87	8
Unsafe	1	7	92

C. Comparative Analysis

Table IV benchmarks the proposed framework against existing approaches in the literature. The proposed system is the only approach to combine WHO-based nutritional validation, visual quality analysis, and regulatory ingredient checking within a unified, interpretable pipeline, achieving the highest classification accuracy of **88.4%**.

TABLE IV
COMPARISON WITH EXISTING APPROACHES

Approach	WHO	Visual	Reg.	Acc.
Calorie Apps	X	X	X	~70%
OCR-Only	✓	X	X	~75%
CNN-Only	X	✓	X	~80%
Rule-based	✓	X	P	~82%

E. Observations and Key Findings

A number of key observations were made during the experimentation process. Firstly, the OCR process was successful in extracting ingredient details from most clear and readable images of food labels. Secondly, the NLP-based preprocessing and normalization of ingredients helped in improving the ingredient mapping process by minimizing spelling variations and synonym mismatches, thus improving the accuracy of ingredient matching. Thirdly, the CNN-based image analysis module was successful in identifying visible contaminants like mold on fruits and discoloration in cooked food. In addition, the WHO rule-based validation module helped in minimizing false positives and improving the accuracy of the system's safety predictions.

VII. LIMITATIONS

Although the system performed well, a number of limitations were observed:

VIII. CONCLUSION AND FUTURE WORK

This paper introduced an AI-based framework for automating food safety and nutrition validation. Its goal is to help consumers make informed dietary choices. The rise in packaged, processed, and fast food consumption, along with complicated food labels and a lack of nutritional information for freshly prepared foods, makes it hard for consumers to find healthier options. To tackle these issues, the proposed system combines optical character recognition, natural language processing, and computer vision techniques to examine both packaged and non-packaged food items in one system.

The framework uses a two-phase design that assesses food label information and actual food images based on World Health Organization (WHO) nutrient guidelines and specific banned ingredient rules in different countries. It extracts ingredient lists, checks nutritional values, identifies visual quality problems, and offers clear safety classifications. This system improves transparency and usability in food safety evaluations. The approach uses rule-based validation and interpretable machine learning methods to ensure the results are clear and dependable for health-related decisions.

The solution shows that **explainable artificial intelligence** can be effectively used in food safety and nutritional analysis. By providing easy-to-understand health reports and safety indicators, the framework aims to raise food awareness and encourage healthier eating habits.

D. OCR Performance

The OCR module performed well in extracting ingredient and nutritional information text from the images of packaged food items. The system performed with an approximate **OCR accuracy of 90-92%**, and the ingredient matching accuracy was approximately **88%** after text normalization. The majority of the errors in the OCR extraction module were attributed to real-world packaging limitations such as extremely small font sizes, curved surfaces of the food labels, and low illumination levels.

